Liquidity Premium Around FOMC Announcements

Tommaso Baglioni* Alessandro Giannozzi† Ruy M. Ribeiro† Oliviero Roggi‡

First version: July 2022

This version: July 2026

Abstract

This paper investigates the relationship between market liquidity and equity returns around scheduled Federal Open Market Committee (FOMC) meetings. We document lower liquidity levels on the day preceding the announcement, with substantial variation across meetings, even in the highly liquid E-mini S&P 500 futures market. We show that ex-ante excess liquidity contains substantial predictive power for post-announcement returns, which have traditionally proven difficult to forecast. Our findings are consistent with a time-varying liquidity premium: periods of low pre-announcement liquidity are associated with lower pre-announcement returns and higher post-announcement returns. We show that this liquidity channel remains a robust determinant of FOMC returns even after controlling for proxies for risk aversion, investor positioning, and monetary policy shocks.

Keywords: FOMC Announcements; Liquidity Premium; Post-FOMC Announcement Return.

JEL Classification: G12; G14.

* Insper, Rua Quatá, 300 – Vila Olímpia, São Paulo – SP, 04546-042, Brazil.

† University of Florence, Via delle Pandette, 32, Firenze (FI), 50127, Italy.

‡ University of Catania, Corso Italia, 55, Catania (CT), 95129, Italy.

1 Introduction

The relation between central bank policies and asset prices is a widely discussed topic in financial economics and has received a lot of attention both in the academic literature and among practitioners, as monetary policy announcements have a substantial impact on the stock market. A large body of literature studies equity return dynamics around Federal Open Market Committee (FOMC) announcements and focuses mainly on the daily stock market return of the FOMC day and on the 24-hour return before scheduled FOMC announcements.[1] Yet, only a handful of papers find interesting patterns related to the stock market behaviour from the monetary policy announcement (or just before) until the end of the trading day.[2] While FOMC announcements cause substantial spikes in equity trading volume (e.g., Rosa, 2016a,b; Bollerslev et al., 2018; Zhu, 2021), the reaction of the stock market is mainly generated by unexpected monetary policy actions (Bernanke and Kuttner, 2005). The average post-announcement return is approximately zero (Lucca and Moench, 2015), and the post-announcement return-to-variance ratio is far exceeded by the pre-announcement counterpart, suggesting that distinct types of risk premia are realized around the announcement window (Hu et al., 2022). A small literature examines the behavior of stock market liquidity around U.S. monetary policy announcements (e.g., Chung et al., 2013; Rosa, 2016a; Karolyi et al., 2019; Lagos and Zhang, 2020). This literature has not studied how liquidity varies across FOMC meetings or whether this variation helps explain returns around the announcements. We address this gap by examining whether market liquidity measured before the announcement contains information about pre- and post-announcement returns and whether FOMC meetings are characterized by a time-varying liquidity premium, defined as the compensation investors require for bearing illiquidity risk.

The main finding of this paper is that market liquidity on the day before scheduled FOMC

[1] The day of the release of the policy statement of a regularly scheduled FOMC meeting is defined henceforth as "event," or "FOMC day." We denote by $\text{FOMC}_{\text{t-1}}$ the trading day before an FOMC day. We refer to "pre-announcement," as the interval that spans from the closing of $\text{FOMC}_{\text{t-1}}$ to the end of the minute immediately preceding the announcement. Similarly, "post-announcement" indicates the interval spanning from the start of the announcement minute to the market close on the FOMC day. When referring to related studies, the expression "post-announcement return" may correspond to a slightly different interval, such as a window around the announcement, but the interpretation of the results is not affected.

[2] Gorodnichenko and Weber (2016) document that most monetary policy shocks within narrow event windows around FOMC decisions are tiny, in line with Neuhierl and Weber (2019), according to which the FOMC releases most of the news about monetary policy outside of scheduled FOMC meetings throughout the year.

announcements varies substantially across meetings and contains information that helps predict post-announcement returns, which have proven difficult to forecast in the existing literature. Rather than documenting a new anomaly, our objective is to examine the predictability of post-announcement returns and their relationship with market liquidity measured before the announcement. Lucca and Moench (2015) show that trading volume and realized volatility are, on average, subdued during the pre-announcement interval and tend to spike at the time of the announcement. This pattern is consistent with the "calm-before-the-storm" effect documented for volatility by Jones et al. (1998) for Treasury securities ahead of inflation and labor-market releases and by Bomfim (2003) for FOMC announcements.

We revisit the calm-before-the-storm effect by focusing on time-series variation in market liquidity and its implications for asset prices. In our framework, the calm-before-the-storm reflects periods of lower trading activity rather than lower stock-return volatility. We show that this liquidity effect is independent of volatility dynamics and we argue that it is reflected in asset prices through a time-varying liquidity premium. Interpreting the calm period as one of reduced trading activity and lower liquidity, we find that the calmer the market before the announcement, the larger the subsequent increase in liquidity and the higher the post-announcement return. These findings suggest that the calm-before-the-storm effect influences asset prices through a liquidity channel.

We focus on FOMC days because monetary policy announcements have a particularly strong impact on asset prices (Savor and Wilson, 2013). Our conjecture is that, although the E-mini S&P 500 futures market is highly liquid, liquidity conditions vary substantially across FOMC meetings and become more important around these announcements. We therefore examine whether liquidity is lower, and its adverse realizations more severe, on the day before FOMC announcements than on non-FOMC days, whether liquidity measured before the announcement helps explain pre- and post-announcement returns, and whether its relation with post-announcement returns remains after controlling for risk aversion, investor positioning, and monetary policy shocks. We do not assume that the liquidity premium is absent on non-FOMC days. Rather, it may be small and difficult to identify on average days, while FOMC uncertainty may make it larger and easier to detect.

The mechanism behind our conjecture is a time-varying liquidity premium. Following the intuition in Amihud (2002), a deterioration in liquidity can cause an immediate negative price adjustment as investors demand greater compensation for holding the asset, while the resulting lower price implies higher expected returns thereafter. If liquidity subsequently recovers,

the required premium falls and prices rise. Because these effects occur at different horizons, the relation between liquidity and pre-announcement returns does not have an unambiguous predicted sign: returns may be negative if liquidity continues to deteriorate, or positive if the compensation for illiquidity or the recovery in liquidity dominates. By contrast, lower liquidity before the announcement followed by a recovery in liquidity should be associated with higher post-announcement returns.

Pre-announcement liquidity may move with broader market conditions, such as risk aversion, uncertainty, or investor positioning ahead of FOMC meetings. Our specifications therefore control directly for observable measures of these conditions, including monetary policy uncertainty and lagged observations of VIX, volatility, and returns, as well as pre-announcement returns, which may capture changes in investor positioning or sentiment before the meeting. The predictive relation between liquidity and post-announcement returns remains robust with these controls in place. As a complementary presentation of the same evidence, the Internet Appendix decomposes excess liquidity into the component explained by these predetermined variables and a residual component. The residual specification is numerically equivalent to including excess liquidity directly alongside the controls; we report it because it shows how much of the variation in pre-FOMC liquidity is explained by prevailing market conditions and how the return relation reads once that component is stripped out. This evidence suggests that the liquidity channel is not merely a proxy for standard measures of risk aversion, uncertainty, or positioning. Instead, liquidity contains distinct information about the frictions faced by market participants ahead of policy announcements, consistent with the compensation required by market makers and arbitrageurs for bearing inventory and asymmetric-information risks.

To capture the multi-dimensional nature of market liquidity, we employ four proxies, described in detail in Section 2.2: market depth, dollar volume, the Amihud (2002) price-impact measure, and the bid-ask spread. We report results for all four measures throughout the paper. Depth and volume directly reflect the trading capacity and market participation central to liquidity provision around scheduled announcements, while the Amihud measure and the bid-ask spread capture price impact and transaction costs. Each proxy is adjusted to reflect excess liquidity by scaling daily liquidity by its trailing monthly average, and we classify an FOMC announcement as preceded by high (low) liquidity if excess liquidity on the day prior to the announcement is positive (not positive). We measure liquidity on the day prior to the announcement so that it is fully predetermined with respect to announcement-day returns;

Section 2.3 discusses this timing choice in detail.[3]

Our study contributes to the literature on equity return dynamics around FOMC announcements by providing an integrated analysis of the relationship between equity market liquidity and returns around scheduled FOMC meetings. The empirical analysis is based on an event study of 215 scheduled FOMC announcements over the period 1997–2024. We use intraday data on E-mini S&P 500 futures at one-minute frequency to construct measures of market liquidity as well as pre- and post-announcement returns.

First, to study the behavior of market liquidity around FOMC announcements, we compare liquidity conditions on the day preceding the announcement with those on non-announcement days. We find that liquidity is lower on $\text{FOMC}_{\text{t-1}}$ days and that its adverse realizations are more severe: the level of liquidity falls, and the lower tail of its distribution becomes fatter. As a consequence of this heterogeneity in pre-announcement market conditions across events, investors may require a more strongly time-varying liquidity premium to compensate for bearing illiquidity risk.

Second, we examine the relationship between ex-ante excess liquidity and intraday returns on both FOMC and non-FOMC days. On FOMC days, ex-ante excess liquidity is positively related to pre-announcement returns and negatively related to post-announcement returns. On non-FOMC days, the estimated relationships are generally negative, with some estimates statistically distinguishable from zero. The relationship between liquidity and afternoon returns is substantially stronger on FOMC days, while the morning-return relationship shifts from generally negative on non-FOMC days to positive on FOMC days. Overall, these findings indicate that the relationship between liquidity and returns varies across FOMC and non-FOMC days and is more pronounced around monetary policy announcements.

Finally, we study post-announcement changes in liquidity and volatility conditional on liquidity on the day before the announcement. We show that liquidity and volatility have distinctive behavior around FOMC announcements. Although both variables spike after the

[3] A small literature documents that FOMC announcements are associated with substantial and systematic intraday fluctuations in liquidity and trading activity. Chung et al. (2013) show that several measures of market liquidity decline prior to scheduled monetary policy announcements and change sharply following the release, consistent with shifts in information flow and trading behavior. Rosa (2016a) documents a deterioration in market quality around FOMC announcements, with liquidity measures worsening immediately after the release, reflecting heightened adverse selection and uncertainty. Focusing on trading activity, Zhu (2021) shows that volume decreases ahead of FOMC announcements and increases sharply afterward, consistent with a temporary withdrawal of trading prior to the release and a resumption of activity once information becomes public.

announcement, low-liquidity meetings exhibit significantly larger increases in liquidity, while volatility responses are similar across liquidity states and not statistically different. Taken together, our findings add to the "calm-before-the-storm" literature by showing that this mechanism is not only reflected in volatility dynamics. We show that there are particular patterns in market liquidity that also have an impact on equity prices.

The findings of this study are relevant for policymakers, portfolio managers, and investors interested in understanding how equity markets respond to monetary policy announcements. Our results are robust to alternative constructions of excess liquidity measures. The remainder of this paper proceeds as follows. Section 2 describes the data and the liquidity measures. The results of the empirical analysis are presented in Section 3. Section 4 reports the robustness tests, and conclusions are offered in Section 5.

2 Data Description and Liquidity Measures

2.1 Data Description

To explore the presence of a liquidity premium around FOMC announcements, we rely on intraday data on E-mini S&P 500 futures at one-minute frequency, which we obtained from Tickdatamarket. Throughout the analysis, we focus on the front contract, the contract closest to expiration, which concentrates the vast majority of trading activity and order-book depth. All prices, volumes, and order-book variables used below therefore refer to the front contract of the E-mini.

Two features of this contract motivate the choice. First, it is the venue where market-wide positions are most cheaply established and unwound. A single transaction provides full index exposure, with low margin requirements and nearly 24-hour trading on Globex. Since 2010, the E-mini alone has out-traded all 500 index constituent stocks combined. Based on our intraday data for the E-mini and on CRSP daily data with the S&P 500 constituent list at each point in time, its full-sample average daily notional trading volume of $137 billion exceeds the constituents' combined $115 billion, with median daily ratios of 1.34 in 2010–2019 and 1.40 in 2020–2024, against 0.72 in 2000–2009. Second, a dominant share of price discovery for the S&P 500 occurs in the E-mini market (Hasbrouck, 2003), continuing a pattern documented for index futures before the E-mini's introduction (Kawaller et al., 1987; Stoll and Whaley, 1990).

The time period of the empirical analysis spans from September 10, 1997 to December

31, 2024.[4] Over this period, there have been a total of 218 scheduled FOMC meetings (eight regularly scheduled meetings per year).[5] Right after the conclusion of every meeting, a policy statement is released by the Federal Reserve. Until March 2011, policy statements are released, on average, at about 2:15 p.m. ET; we obtained the exact release time from the Internet Appendix of Lucca and Moench (2015).[6] Since April 2011, policy statements have been released at 2:00 p.m., except for eight meetings when the policy statement was released at 12:30 p.m. From the website of the Federal Reserve, we obtained data and information on FOMC meeting calendars, Federal funds rate, and monetary policy decisions.[7]

To mitigate data errors and outliers, we apply a series of standard data-cleaning filters. Specifically, we verify that bid and ask prices, bid and ask sizes, bid-ask spreads, and trading volume are non-negative and non-missing; that mid-prices are within a reasonable range relative to the S&P 500 Index; and that bid-ask spreads are lower than four index points.

2.2 Measuring Market Liquidity

As proxies for market liquidity, we use four measures that capture different dimensions of liquidity, and we report results for all four measures throughout the paper. Market depth and dollar volume directly capture trading capacity and market participation, which are central to our interpretation of liquidity provision around scheduled monetary policy announcements. Depth reflects the quantity that can be traded without moving prices, while volume measures the extent of actual trading activity, making both measures particularly well suited to capturing the strategic withdrawal and subsequent re-entry of liquidity around FOMC meetings.[8]

[4] The FOMC started announcing the outcome of the meetings in February 1994. In February 1995, the FOMC announced that all monetary policy changes would be communicated to the public immediately. In January 2000, the FOMC announced that, following each regularly scheduled meeting, it would issue a statement even if there had been no change in monetary policy. Our analysis starts when E-mini S&P 500 was launched.

[5] As for two FOMC meetings we do not have complete intraday data on the volume, we exclude those two meetings. As Baglioni and Ribeiro (2023), we exclude the FOMC meeting of October 2008, which is characterized by a 24-hour pre-announcement return greater than 10% (extreme observation). We thus consider 215 events in our empirical analysis.

[6] All times mentioned in this paper are in Eastern Time (ET).

[7] Detailed information on FOMC meeting calendars and statements is available at `https://www.federalreserve.gov/monetarypolicy/fomccalendars.htm`.

[8] Our dataset reports the best bid and ask quotes only, so depth is measured at the top of the book. Deeper levels of the order book, as well as hidden or iceberg liquidity, are not observable in our data.

The Amihud (2002) measure captures the price response associated with one dollar of trading volume, thus serving as a measure of price impact. Bid-ask spread reflects the size of the transaction costs.

We deliberately measure liquidity in the futures order book rather than through aggregates of individual-stock liquidity; Section 4 discusses this choice and reports that the relation we document does not carry over to liquidity aggregates of the S&P 500 constituent stocks.

We measure market liquidity on trading day t and minute m ($Liquidity_{t,m}$), using the following liquidity measures:

$$Depth_{t,m} = Mid\ Price_{t,m} \times [Ask\ Size_{t,m} + Bid\ Size_{t,m}]/2 \tag{1}$$

$$Dollar\ Volume_{t,m} = 50 \times Mid\ Price_{t,m} \times Volume_{t,m} \tag{2}$$

$$Amihud\ Liquidity_{t,m} = [\frac{|R_{t,m}|}{Dollar\ Volume_{t,m}}]^{-1} \tag{3}$$

$$Spread\ Tightness_{t,m} = [Ask\ Price_{t,m} - Bid\ Price_{t,m}]^{-1} \tag{4}$$

where Ask (Bid) $Price_{t,m}$ represents the Ask (Bid) closing price of minute m. Ask (Bid) $Size_{t,m}$ represents the number of contracts that investors are trying to buy at the Ask (Bid) $Price_{t,m}$. $Mid\ Price_{t,m}$ indicates the mid-price closing value of minute m, and is equal to $0.5 \times (Ask\ Price_{t,m} + Bid\ Price_{t,m})$. $|R_{t,m}|$ is the absolute value of the mid-price 1-minute return. $Dollar\ Volume_{t,m}$ is computed as the product of the number of contracts traded during one minute ($Volume_{t,m}$) and the value paid to enter one E-mini S&P 500 contract at the end of that minute ($50 \times Mid\ Price_{t,m}$). The liquidity measures are computed using only regular stock market trading hours.

For each measure of liquidity, we compute daily excess liquidity as follows: first, for every minute of every trading day, we compute liquidity as in equations (1) to (4).[9] Second, we take the average of all minutes for each trading day (daily liquidity). Third, we divide the daily liquidity of each trading day by its daily average of the previous month. By controlling for the average liquidity of the previous month, we make sure that we are using a proxy for the liquidity that is unexpected (or out of the ordinary), reflecting innovations in liquidity.

In equations (3) and (4), we take the reciprocal to reflect liquidity instead of illiquidity.

[9] We include only minute-level observations from regular stock market trading hours (9:30 a.m. to 4:00 p.m.).

Intuitively, the higher the price impact, the higher the illiquidity; as we take the reciprocal, the higher the value of the Amihud measure, the higher the liquidity (instead of the illiquidity) of the market. The same applies to the bid-ask spread, which reflects the size of the transaction costs: its reciprocal reflects liquidity instead of illiquidity, and we refer to it as spread tightness.[10]

Table A.4 reports the correlation among our liquidity measures on $FOMC_{t-1}$ and non-FOMC days (defined as trading days different from $FOMC_{t-1}$, $FOMC_t$, and $FOMC_{t+1}$). As shown in Table A.4, the correlation across the four variables tends to be stronger on the day before FOMC announcements than on non-FOMC days.

2.3 Variables Description

The main variables in the paper, together with the liquidity measures described in the previous section, are the pre- and post-announcement returns. We compute these returns using minute-level midpoint prices for E-mini S&P 500 futures. The pre-announcement return is measured from the close of trading on $FOMC_{t-1}$ (3:59 p.m.) to the end of the minute immediately preceding the announcement. The post-announcement return is measured from the beginning of the announcement minute to the close of trading on the FOMC day (3:59 p.m.). In regressions of post-announcement returns and in analyses that condition these returns on the level of prior market performance, we use the return over the 24 hours preceding the announcement.

For non-FOMC days, we use morning and afternoon returns as counterparts to the pre- and post-announcement returns, respectively. The morning return is measured from the close of the previous trading day to 1:59 p.m., whereas the afternoon return is measured from 2:00 p.m. to 3:59 p.m. All returns are expressed in basis points.

We distinguish between announcements characterized by high and low liquidity. We define an FOMC announcement as associated with a high (low) level of liquidity if liquidity on $FOMC_{t-1}$ is higher (not higher) than the average daily liquidity of the previous month (i.e., if excess liquidity is either positive or negative). We measure market liquidity on the trading day preceding the FOMC announcement to ensure that liquidity is fully predetermined with respect to announcement-day returns and to avoid concerns arising from the sharp intraday variation in trading activity observed on FOMC days. In fact, equity market liquidity on announcement days exhibits pronounced intraday swings and discrete jumps at the time of the

[10] For a comprehensive review of the literature on liquidity measures, see Le and Gregoriou (2020).

announcement (Chung et al., 2013), making daily or intraday measures difficult to interpret and mechanically related to contemporaneous price changes. By contrast, liquidity measured on the previous trading day captures market conditions entering the announcement and reflects investors' willingness to supply liquidity ahead of a known information event.

We identify Federal funds rate (FFR) target and path shocks using data from Acosta et al. (2024), which updated previous data from Nakamura and Steinsson (2018).[11] As a measure of monetary policy shocks, we also use the first principal component of high-frequency surprises in short-rate derivatives (JK shock) from Jarociński and Karadi (2020).[12] As a proxy for volatility, we use the difference between the high and low mid-price values within each minute, divided by the closing mid-price for that minute. We compute daily volatility, as the average volatility across all minutes of the trading day. Along with volatility, as control variables, we use the open to close return of the previous trading day and the Monetary Policy Uncertainty (MPU) Index from Husted et al. (2020).[13]

3 Empirical Findings

This section presents our empirical findings. We first examine the time-series properties of excess liquidity. We then analyze the relation between ex-ante excess liquidity and intraday returns on FOMC and non-FOMC days, assessing whether the relation differs across the two settings and is consistent with a liquidity premium. Finally, we analyze post-announcement changes in liquidity and volatility as a function of pre-announcement liquidity.

[11] The FFR target shock captures the unexpected component of the change in the current Federal funds rate. The FFR path shock captures revisions to expectations about the future path of monetary policy conveyed by the FOMC announcement, over and above the surprise associated with the current target-rate decision. Data is available at `https://www.acostamiguel.com/research.html`.

[12] Data is available at `https://github.com/marekjarocinski`.

[13] The MPU Index is a measure of the uncertainty that the public perceives about Federal Reserve monetary policy actions and their consequences. It is constructed by tracking the frequency of newspaper articles related to monetary policy uncertainty, searching for articles in the Washington Post, Wall Street Journal, and New York Times, containing the triple of "uncertainty" or "uncertain;" "monetary policy(ies)" or "interest rate(s)" or "Federal fund(s) rate" or "Fed fund(s) rate;" and "Federal Reserve" or "the Fed" or "Federal Open Market Committee" or "FOMC." Data on the MPU Index are available at `https://sites.google.com/site/lucasfhusted/data`.

3.1 The Variation in Market Liquidity Around FOMC Announcements

We begin by examining the behavior of our liquidity measures around FOMC announcements. Specifically, we ask how liquidity conditions on $\text{FOMC}_{\text{t-1}}$ days differ from those on non-FOMC days, in level and in the severity of their lower tail.

Table 1 provides evidence that liquidity conditions deteriorate before FOMC announcements, although the nature of the deterioration differs across measures. For depth and volume, the entire center of the distribution shifts downward. Median excess depth and volume are -7.0% and -6.7%, respectively, on $\text{FOMC}_{\text{t-1}}$ days, compared with 1.3% and 0.8% on non-FOMC days. The corresponding differences of -8.3 and -7.4 percentage points are statistically significant. For Amihud and spread liquidity, median differences are small and statistically insignificant. However, the lower tails of their distributions are substantially more negative before FOMC announcements. At the 10th percentile, Amihud liquidity is -49.8% on $\text{FOMC}_{\text{t-1}}$ days, compared with -27.4% on non-FOMC days, a difference of -22.4 percentage points. The corresponding spread-liquidity values are -2.57% and -1.48%, a difference of -1.09 percentage points. Thus, Amihud and spread liquidity do not exhibit a systematic decline on the typical pre-FOMC day, but particularly adverse liquidity realizations are more severe. The same asymmetry appears in the frequency of adverse realizations: only 39.5% of $\text{FOMC}_{\text{t-1}}$ days have positive excess depth, compared with 52.1% of non-FOMC days.

INSERT TABLE 1

Figures A.1–A.3 provide graphical evidence on the level, classification, and dispersion of excess liquidity around FOMC announcements. Figure A.1 is the graphical counterpart to Table 1, comparing excess liquidity on $\text{FOMC}_{\text{t-1}}$ and non-FOMC days. Figure A.2 reports the shares of high- and low-liquidity observations throughout the event window, showing how frequently liquidity is above or at or below its prior-month benchmark. Figure A.3 presents the interdecile range of excess liquidity from one week before the announcement through the pre- and post-announcement intervals on the announcement day. By displaying the 10th and 90th percentiles together with the median, it illustrates how both the location and dispersion of liquidity evolve around FOMC announcements. Together, the figures complement Table 1 by showing that pre-FOMC liquidity conditions differ not only in their typical level but also in the frequency and severity of low-liquidity realizations.

The results from this section suggest that the day before FOMC announcements is characterized by abnormally low levels of liquidity. Due to the cross-event heterogeneity in pre-announcement market conditions, investors might require a liquidity premium as compensation for bearing the risk of investing in the interval before and/or after the FOMC announcement.

3.2 Excess Liquidity and Intraday Returns on FOMC and Non-FOMC Days

In the previous section, we document abnormally low liquidity levels and more severe adverse liquidity realizations on $\text{FOMC}_{\text{t-1}}$ days. In this section, we examine whether ex-ante excess liquidity predicts intraday market returns and whether this relationship differs between FOMC and non-FOMC days. For each liquidity measure, we estimate the following regressions:

$$\begin{aligned} Afternoon\ Return_t = \alpha + \theta FOMC_t + \beta \ln(1 + Excess\ Liquidity_{t-1}) \\ + \phi \left[\ln(1 + Excess\ Liquidity_{t-1}) \times FOMC_t\right] + \delta' X_t + \varepsilon_t, \end{aligned} \tag{5}$$

$$\begin{aligned} Morning\ Return_t = \alpha + \theta FOMC_t + \beta \ln(1 + Excess\ Liquidity_{t-1}) \\ + \phi \left[\ln(1 + Excess\ Liquidity_{t-1}) \times FOMC_t\right] + \delta' X_t + \varepsilon_t. \end{aligned} \tag{6}$$

where $FOMC_t$ is an indicator variable equal to one if day t is a scheduled FOMC announcement day and zero otherwise. The omitted group consists of designated non-FOMC days. The coefficient θ represents the incremental FOMC-day intercept shift at this benchmark level of liquidity, holding the remaining covariates constant. The coefficient β measures the slope of the relationship between lagged excess liquidity and returns on non-FOMC days, while $\beta + \phi$ measures the corresponding slope on FOMC days. The interaction coefficient ϕ directly captures the difference between the FOMC and non-FOMC slopes. In equation (5), a negative and statistically significant ϕ indicates that the relationship between lagged liquidity and post-announcement returns is more negative on FOMC days than on non-FOMC days. A negative and statistically significant ϕ, together with a negative $\beta + \phi$, indicates that lower lagged liquidity predicts higher post-announcement returns on FOMC days. Analogously, in equation (6), a positive and statistically significant ϕ, together with a positive $\beta + \phi$, indicates that the relationship between lagged liquidity and pre-announcement returns is more positive on FOMC days. The vector X_t includes lagged daily volatility and weekday, month, and year fixed effects. Equation (5) additionally controls for the previous trading day's afternoon return, whereas equation (6) controls for the previous trading day's open-to-close return. We estimate both

equations using daily data over the 1997–2024 sample period. Statistical inference is based on Newey–West standard errors.

On FOMC days, morning and afternoon returns correspond, respectively, to the pre- and post-announcement returns; Sections 2.2 and 2.3 provide detailed definitions of the return windows and of the remaining variables in equations (5) and (6).

Table 2 reports the relationship between lagged excess liquidity and afternoon returns. On FOMC announcement days, the liquidity slopes are negative and statistically significant across all four measures, indicating that greater liquidity on the preceding trading day is associated with lower post-announcement returns. The contrast with designated non-FOMC days is especially pronounced for Amihud liquidity and spread tightness, whose non-FOMC slopes are close to zero. Although depth and volume are also negatively related to afternoon returns on non-FOMC days, their FOMC-day slopes are considerably larger in absolute magnitude. Formal tests show that, for each measure, the FOMC-day liquidity slope is significantly more negative than its non-FOMC counterpart.

The FOMC-day intercept shifts are statistically insignificant in all four specifications. Thus, when prior-day liquidity equals its trailing-month benchmark, there is no detectable fixed difference between FOMC and non-FOMC afternoon returns, conditional on the controls and fixed effects. Instead, the distinction between the two types of days arises from how afternoon returns vary with prior liquidity conditions. This pattern is consistent with Savor and Wilson (2013), who document that the risk–return relationship differs on days with scheduled macroeconomic announcements. Our findings extend this evidence to market liquidity, indicating that scheduled monetary-policy announcements alter the relationship between ex-ante liquidity and subsequent returns rather than simply producing a uniform shift in average afternoon returns.[14]

INSERT TABLE 2

Table A.6 reports the corresponding results for morning returns. The FOMC-day liquidity slopes are positive and statistically significant for depth and volume, but not for the Amihud or spread measures. The FOMC-minus-non-FOMC slope differences are also positive and significant for depth and volume. The FOMC-day intercept shift is positive and statistically significant across all four specifications, indicating that, when prior-day liquidity equals its

[14] The conclusions are robust to controlling for lagged VIX levels or unexpected volatility rather than realized volatility.

trailing-month benchmark, conditional morning returns are higher on FOMC days than on designated non-FOMC days.

Tables 2 and A.6 together reveal an asymmetric relationship between ex-ante liquidity and the timing of returns around FOMC announcements. Higher excess depth and volume are associated with larger morning returns but lower afternoon returns on FOMC days. For Amihud liquidity and spread tightness, the evidence is concentrated in the afternoon-return regressions.

The negative relation between ex-ante excess liquidity and subsequent market returns reported in Table 2, and confirmed in Table 3 below, is consistent with the broader literature on liquidity and asset prices. This literature shows that liquidity predicts stock returns at both the firm and aggregate market levels. Amihud and Mendelson (1986a,b) were the first to examine the effect of bid–ask spreads on security returns. They show that expected stock returns increase with illiquidity: the larger a stock's spread, the higher its expected return. Subsequent contributions include Amihud and Mendelson (1989), Brennan and Subrahmanyam (1996), Brennan et al. (1998), Datar et al. (1998), Easley et al. (2002), Baker and Stein (2004), Acharya and Pedersen (2005), Lee (2011), and Kim and Lee (2014). At the aggregate level, Amihud (2002) shows that ex-ante excess stock returns increase with expected market illiquidity, whereas unexpected illiquidity is negatively related to contemporaneous returns. The latter relation arises because an unexpected increase in illiquidity raises required returns and therefore reduces current stock prices.

A related theoretical literature examines how market-wide events affect stock liquidity. These studies predict that the demand for liquidity increases following large market declines as investors liquidate positions across assets, while the supply of liquidity contracts as liquidity providers face binding wealth or funding constraints.[15] Consistent with these predictions, Hameed et al. (2010) and Karolyi et al. (2012) empirically document that stock illiquidity increases following negative market returns, particularly when funding conditions are tight. Amihud et al. (2006) provide a comprehensive review of the theoretical and empirical literature on liquidity and asset prices.

[15] The main contributions include the contagion models of Kyle and Xiong (2001), Pavlova and Rigobon (2008), and Gromb and Vayanos (2002, 2018); the coordination-failure models of Bernardo and Welch (2004), Morris and Shin (2004), and Vayanos (2004); and the collateral-based models of Anshuman and Viswanathan (2006) and Brunnermeier and Pedersen (2009).

3.3 Excess Liquidity and Intraday Returns on FOMC Days Only

We next examine the relation between ex-ante excess liquidity and intraday returns on FOMC days in greater detail, controlling for other factors that may explain returns around FOMC announcements. We estimate the following regressions:

$$\textit{Post-Ann. Return}_t = \alpha + \beta \ln(1 + \textit{Excess Liquidity}_{t-1}) + \gamma\, FFR\ Target\ Shock_t + \eta\, FFR\ Path\ Shock_t + \delta' X_t + \varepsilon_t \quad (7)$$

$$\textit{Pre-Ann. Return}_t = \alpha + \beta \ln(1 + \textit{Excess Liquidity}_{t-1}) + \delta' X_t + \varepsilon_t \quad (8)$$

where the FFR target and path shocks capture unexpected changes in the current Federal funds rate and its expected future path, respectively. The control variables (X_t) include volatility on $\text{FOMC}_{\text{t-1}}$ and the level of the MPU Index associated with each event. Equation (7) also controls for the 24-hour return preceding the announcement, whereas equation (8) controls for the open-to-close return on $\text{FOMC}_{\text{t-1}}$. These variables are intended to capture the potential effects of risk aversion and investor positioning before the meeting. The sample consists exclusively of FOMC announcement days from 1997 through 2024.[16]

INSERT TABLE 3

Table 3 shows a negative and statistically significant relation between ex-ante excess liquidity and post-announcement returns on FOMC days. This relation remains significant after controlling for other potential determinants of equity returns following FOMC announcements, including Federal funds rate target and path shocks and the 24-hour return preceding the announcement. The Internet Appendix (Section 7.9) re-estimates this specification excluding the eight meetings with a 12:30 p.m. release and, separately, including the October 2008 meeting.

In contrast, Table A.7 shows a positive estimated relationship between ex-ante excess liquidity and pre-announcement returns across all four liquidity measures, although the evidence is statistically significant only for depth and Amihud liquidity. Lagged daily volatility is positively and strongly related to pre-announcement returns, consistent with Martello and Ribeiro (2018) and Neuhierl and Weber (2019), who find that the pre-FOMC announcement drift is

[16] We estimate equations (7) and (8) using 210 events rather than 215 because data on Federal funds rate shocks and the MPU Index are available only through the first half of 2024. Results based on the full sample of 215 events are reported in Table A.5. The corresponding specifications control for the first principal component of high-frequency surprises in short-rate derivatives from Jarociński and Karadi (2020) and yield qualitatively similar results.

concentrated in periods of elevated market uncertainty. By contrast, the MPU Index enters negatively, suggesting that realized market volatility and event-specific monetary-policy uncertainty capture distinct dimensions of the pre-announcement environment.

Taken together, these results are consistent with the presence of a liquidity premium around FOMC meetings. In particular, the positive relation between excess liquidity and pre-announcement returns, combined with its negative relation with post-announcement returns, suggests that compensation for bearing illiquidity risk is reflected in the pattern of returns around FOMC announcements.

Taken literally, the estimates in Table 3 imply that meetings preceded by high liquidity earn a negative return after the announcement, which would be an implausible negative risk premium. Across all meetings, the average pre-announcement return is 27.75 basis points ($t = 5.72$), while the average post-announcement return is indistinguishable from zero; a negative post-announcement average after high-liquidity days therefore says little, by itself, about the premium those meetings pay in total. Table 4 is designed to interpret the magnitudes during the full announcement date. The dependent variable is the full announcement-date return, from the close of $\text{FOMC}_{\text{t-1}}$ to the close of the FOMC day, so that the return delivered before and after the release is measured jointly; the regression controls for the meeting's own pre-announcement return; and the constant is replaced by liquidity-state dummies that sum to one, so that each coefficient measures the fitted announcement-day return in its liquidity state.

INSERT TABLE 4

Read this way, the magnitudes are still puzzling, but we connect two potential puzzles: the pre-announcement drift and the liquidity relationship. After low-liquidity days, the announcement-day return averages 32 to 39 basis points, significant at the 1% level; after high-liquidity days, it averages 4 to 17 basis points, generally indistinguishable from zero. The low–high differential is estimated with less precision than the state levels themselves: it is significant for depth (33 basis points) and of the same sign but insignificant for the other three measures, so the evidence speaks more clearly to where the return accrues than to how much larger it is in low-liquidity states. The Internet Appendix reports consistent results using the post-announcement return, the adjusted post-announcement return that adds back the average pre-announcement premium, and liquidity terciles (Tables A.10 to A.13). In this description, high liquidity on $\text{FOMC}_{\text{t-1}}$ is not associated with a negative return over the full announcement day, but it is followed, on average, by a negative return in the post-announcement period. The average re-

turn thus appears to accrue mostly in the pre-announcement window, while the variation across meetings that is related to liquidity appears mostly after the announcement.

3.4 Liquidity and Volatility Over the Post-FOMC Announcement Interval

A large literature documents that scheduled macroeconomic announcements are preceded by unusually low trading activity and volatility, followed by sharp increases at the time of the release. Jones et al. (1998) provide early evidence of a "calm-before-the-storm" effect in Treasury markets ahead of inflation and labor market announcements, documenting unusually low trading activity and volatility prior to the release and sharp increases thereafter, consistent with market participants awaiting the resolution of public information. Bomfim (2003) extends this mechanism to monetary policy announcements, showing that volatility is subdued prior to scheduled FOMC meetings and spikes sharply once the policy decision is released. Consistent with this evidence, subsequent works document similar dynamics in asset prices around FOMC announcements. Lucca and Moench (2015), Rosa (2016b), and Bollerslev et al. (2018) show that realized volatility and trading volume increase sharply after FOMC announcements. Similarly, Chung et al. (2013), Rosa (2016a), and Zhu (2021) document substantial increases in equity trading volume following these events.

Building on this literature, we examine the behavior of liquidity and volatility in the post-announcement interval, conditioning on the level of liquidity observed on the day before the announcement. We measure percentage changes in liquidity and volatility as the ratio of their average values over the post-announcement interval to their average values over the final 30 minutes preceding the announcement. Table 5 reports the resulting average spikes in liquidity and volatility across liquidity states.

INSERT TABLE 5

The table shows that liquidity rebounds sharply after FOMC announcements, especially when market liquidity was relatively low on the preceding trading day. The pattern is economically large and robust across both measures of liquidity: post-announcement depth increases by 247% following low-depth days, compared with 205% following high-depth days, while dollar volume increases by 364% following low-volume days, compared with 297% following high-volume days. The low–high differentials, 42 percentage points for depth and 67 percentage points for volume, are statistically significant and remain similar when using winsorized means

or medians. By contrast, although volatility also rises substantially after the announcement, its increase does not differ significantly across the low- and high-liquidity groups. Thus, low pre-announcement liquidity predicts a stronger subsequent recovery in market liquidity but not a significantly larger volatility response, suggesting that the liquidity result is not simply attributable to differences in post-announcement volatility. The Internet Appendix (Section 7.5, Table A.14) reports these post-announcement changes separately by the sign of the pre-announcement return and by the sign of the JK policy shock.

Our findings thus extend the calm-before-the-storm literature by showing that this mechanism, previously documented primarily in terms of volatility, is also closely linked to market liquidity and is priced in returns.

4 Supporting Evidence and Robustness Tests

The Internet Appendix reports supporting evidence and robustness tests for the main results; for convenience, it is presented at the end of the main text. It is organized as follows. First, Appendix 7.1 constructs the same liquidity measures for the S&P 500 constituent stocks and shows that the return relation is specific to the E-mini. Appendix 7.2 reports descriptive statistics for the liquidity measures, including correlations, high- and low-liquidity classifications, and the dispersion of excess liquidity around FOMC announcements. Appendix 7.3 then examines whether the liquidity-return relation is robust to alternative intraday return windows, including specifications based on pre-announcement liquidity measured during the morning window, and to the use of the JK policy shock from Jarociński and Karadi (2020). Appendix 7.4 reports the specification of Table 4 under alternative definitions of the announcement-day return and of the liquidity states. Appendix 7.5 reports the post-announcement liquidity changes separately by the direction of the pre-announcement return and of the policy surprise. Appendix 7.6 presents the relation using residual liquidity, the component of excess liquidity not explained by predetermined market variables. Appendix 7.7 repeats the main specifications over the more recent 2015–2024 subsample, excluding 2020 and 2021 to reduce the influence of COVID-related market outliers. Appendix 7.8 examines how the liquidity premium relates to recent macroeconomic news. Finally, Appendix 7.9 re-estimates the main specification excluding the eight 12:30 p.m. release meetings and, separately, including the October 2008 meeting.

Liquidity of the S&P 500 Constituent Stocks. As anticipated in Section 2.2, we measure liquidity in the futures order book rather than through aggregates of individual-stock liquidity. Market-wide positions are typically not built by trading hundreds of individual stocks, so cross-sectional averages of stock-level liquidity measure a different object: the average cost of trading single names, not the cost of trading index exposure. Individual-stock liquidity does contain a common component (Chordia et al., 2000), and in Appendix 7.1 we show that the same measures constructed from CRSP daily data, using the S&P 500 constituent list at each point in time, correlate positively with our futures measures. The relation we document, however, does not carry over to these aggregates (Tables A.1 to A.3), consistent with the premium being tied to the liquidity of index-exposure trading. An E-mini position can also be established and unwound continuously through the announcement window, including the overnight segment of the pre-announcement interval that begins at the close of $\text{FOMC}_{\text{t-1}}$, when constituent stocks trade thinly at best.

Modifying the Excess Liquidity Measures. The results are robust to alternative constructions of excess liquidity. In particular, we obtain similar findings when liquidity is measured during the pre-announcement interval, on the last trading day of the week preceding the announcement, or on the first trading day of the FOMC week, rather than on $\text{FOMC}_{\text{t-1}}$. The results are also robust to alternative benchmarks used to define excess liquidity: replacing the trailing monthly average with the corresponding median yields qualitatively similar conclusions. In the baseline regression specifications, excess liquidity is log-transformed to reduce the influence of extreme observations. The results are qualitatively unchanged when excess liquidity is included in levels instead.

5 Conclusions

Our study contributes to the literature on equity return dynamics around FOMC announcements by providing an integrated analysis of the relationship between market liquidity and equity returns. While a small number of papers examine stock market liquidity around U.S. monetary policy announcements, they do not investigate the time-series variation in market liquidity across FOMC meetings or its implications for announcement-day returns.

The main finding of this paper is that pre-announcement market liquidity contains useful information for predicting post-announcement returns, a quantity that has proven difficult to

forecast in the existing literature. Interpreting low pre-announcement liquidity as a manifestation of a "calm-before-the-storm" environment, characterized by reduced trading activity and the temporary withdrawal of liquidity provision, we find that calmer pre-announcement periods are followed by larger post-announcement increases in liquidity and higher post-announcement returns. Measured over the full announcement day, from the close of $\text{FOMC}_{\text{t-1}}$ to the close of the FOMC day, meetings preceded by low liquidity deliver 32 to 39 basis points, whereas meetings preceded by high liquidity deliver returns that are generally indistinguishable from zero: high ex-ante liquidity signals that the announcement premium has already been delivered before the release.

Our results extend the calm-before-the-storm story of Jones et al. (1998) and Bomfim (2003) by showing that pre-announcement liquidity conditions are not only associated with subsequent changes in trading activity and volatility, but are also priced in post-announcement returns. While volatility increases following FOMC announcements are similar across liquidity states, liquidity responses differ markedly: post-announcement liquidity increases are significantly larger following low-liquidity meetings. This distinction indicates that our findings are not driven by differential volatility responses.

Our results are subject to a limitation. Excess liquidity is not randomly assigned and may proxy for unmeasured determinants of announcement-day returns, such as dealer inventory, disagreement, or investor positioning; the controls in Table 3 address the measurable candidates but cannot rule out all of them.

Several avenues for future research emerge from this paper. One promising direction is the development of theoretical frameworks that can rationalize the presence of a liquidity premium around scheduled monetary policy announcements. Existing models of return dynamics around FOMC meetings may need to be extended to account for the role of liquidity highlighted in this paper, while some may be challenged by the results that we document.

References

V. V. Acharya and L. H. Pedersen. Asset pricing with liquidity risk. *Journal of Financial Economics*, 77(2):375–410, 2005.

M. Acosta, C. M. Brennan, and M. M. Jacobson. Constructing high-frequency monetary policy surprises from SOFR futures. *Economics Letters*, 242:111873, 2024.

Y. Amihud. Illiquidity and stock returns: cross-section and time-series effects. *Journal of Financial Markets*, 5(1):31–56, 2002.

Y. Amihud and H. Mendelson. Asset pricing and the bid-ask spread. *Journal of Financial Economics*, 17(2):223–249, 1986a.

Y. Amihud and H. Mendelson. Liquidity and Stock Returns. *Financial Analysts Journal*, 42 (3):43–48, 1986b.

Y. Amihud and H. Mendelson. The Effects of Beta, Bid-Ask Spread, Residual Risk, and Size on Stock Returns. *The Journal of Finance*, 44(2):479–486, 1989.

Y. Amihud, H. Mendelson, and L. H. Pedersen. Liquidity and Asset Prices. *Foundations and Trends® in Finance*, 1(4):269–364, 2006.

R. Anshuman and V. R. Viswanathan. Collateral and Market Liquidity. *Working Paper*, 2006.

T. Baglioni and R. Ribeiro. Corporate Bonds Distress and FOMC Announcement Returns. *Working Paper*, 2023.

M. Baker and J. C. Stein. Market liquidity as a sentiment indicator. *Journal of Financial Markets*, 7(3):271–299, 2004.

B. S. Bernanke and K. N. Kuttner. What Explains the Stock Market's Reaction to Federal Reserve Policy? *The Journal of Finance*, 60(3):1221–1257, 2005.

A. E. Bernardo and I. Welch. Liquidity and Financial Market Runs. *The Quarterly Journal of Economics*, 119(1):135–158, 2004.

T. Bollerslev, J. Li, and Y. Xue. Volume, volatility, and public news announcements. *The Review of Economic Studies*, 85(4):2005–2041, 2018.

A. N. Bomfim. Pre-announcement effects, news effects, and volatility: Monetary policy and the stock market. *Journal of Banking Finance*, 27(1):133–151, 2003.

M. J. Brennan and A. Subrahmanyam. Market microstructure and asset pricing: On the compensation for illiquidity in stock returns. *Journal of Financial Economics*, 41(3):441–464, 1996.

M. J. Brennan, T. Chordia, and A. Subrahmanyam. Alternative factor specifications, security characteristics, and the cross-section of expected stock returns. *Journal of Financial Economics*, 49(3):345–373, 1998.

M. K. Brunnermeier and L. H. Pedersen. Market Liquidity and Funding Liquidity. *The Review of Financial Studies*, 22(6):2201–2238, 2009.

T. Chordia, R. Roll, and A. Subrahmanyam. Commonality in liquidity. *Journal of Financial Economics*, 56(1):3–28, 2000.

K. H. Chung, J. Elder, and J.-C. Kim. Liquidity and Information Flow around Monetary Policy Announcement. *Journal of Money, Credit and Banking*, 45(5):781–820, 2013.

V. T. Datar, N. Y. Naik, and R. Radcliffe. Liquidity and stock returns: An alternative test. *Journal of Financial Markets*, 1(2):203–219, 1998.

D. Easley, S. Hvidkjaer, and M. O'Hara. Is Information Risk a Determinant of Asset Returns? *The Journal of Finance*, 57(5):2185–2221, 2002.

Y. Gorodnichenko and M. Weber. Are Sticky Prices Costly? Evidence from the Stock Market. *American Economic Review*, 106(1):165–199, 2016.

D. Gromb and D. Vayanos. Equilibrium and welfare in markets with financially constrained arbitrageurs. *Journal of Financial Economics*, 66(2-3):361–407, 2002.

D. Gromb and D. Vayanos. The Dynamics of Financially Constrained Arbitrage. *The Journal of Finance*, 73(4):1713–1750, 2018.

A. Hameed, W. Kang, and S. Viswanathan. Stock Market Declines and Liquidity. *The Journal of Finance*, 65(1):257–293, 2010.

J. Hasbrouck. Intraday price formation in U.S. equity index markets. *Journal of Finance*, 58 (6):2375–2400, 2003.

G. X. Hu, J. Pan, J. Wang, and H. Zhu. Premium for heightened uncertainty: Explaining pre-announcement market returns. *Journal of Financial Economics*, 145(3):909–936, 2022.

L. Husted, J. Rogers, and B. Sun. Monetary policy uncertainty. *Journal of Monetary Economics*, 115:20–36, 2020.

M. Jarociński and P. Karadi. Pre-announcement effects, news effects, and volatility: Monetary policy and the stock market. *American Economic Journal: Macroeconomics*, 12(2):1–43, 2020.

C. M. Jones, . Lamont, and R. L. Lumsdaine. Macroeconomic news and bond market volatility. *Journal of of Financial Economics*, 47(3):315–337, 1998.

G. A. Karolyi, K.-H. Lee, and M. A. van Dijk. Understanding commonality in liquidity around the world. *Journal of Financial Economics*, 105(1):82–112, 2012.

G. A. Karolyi, K.-H. Lee, and M. A. van Dijk. U.S. monetary policy transmission and liquidity risk premia around the world. *Working Paper*, 2019.

I. G. Kawaller, P. D. Koch, and T. W. Koch. The temporal price relationship between S&P 500 futures and the S&P 500 index. *Journal of Finance*, 42(5):1309–1329, 1987.

S.-H. Kim and K.-H. Lee. Pricing of liquidity risks: Evidence from multiple liquidity measures. *Journal of Empirical Finance*, 25:112–133, 2014.

A. S. Kyle and W. Xiong. Contagion as a Wealth Effect. *The Journal of Finance*, 56(4): 1401–1440, 2001.

R. Lagos and S. Zhang. Turnover Liquidity and the Transmission of Monetary Policy. *American Economic Review*, 110(6):1635–1672, 2020.

H. Le and A. Gregoriou. How do you capture liquidity? A review of the literature on low-frequency stock liquidity. *Journal of Economic Surveys*, 34(5):1170–1186, 2020.

K.-H. Lee. The world price of liquidity risk. *Journal of Financial Economics*, 99(1):136–161, 2011.

D. O. Lucca and E. Moench. The Pre-FOMC Announcement Drift. *The Journal of Finance*, 70(1):329–371, 2015.

V. Martello and R. Ribeiro. Pre-FOMC Announcement Relief. *Working Paper*, 2018.

S. Morris and H. S. Shin. Liquidity Black Holes. *Review of Finance*, 8(1):1–18, 2004.

E. Nakamura and J. Steinsson. High-Frequency Identification of Monetary Non-Neutrality: The Information Effect. *The Quarterly Journal of Economics*, 133(3):1283–1330, 2018.

A. Neuhierl and M. Weber. Monetary policy communication, policy slope, and the stock market. *Journal of Monetary Economics*, 108:140–155, 2019.

A. Pavlova and R. Rigobon. The Role of Portfolio Constraints in the International Propagation of Shocks. *The Review of Economic Studies*, 75(4):1215–1256, 2008.

C. Rosa. Walking on thin ice: Market quality around FOMC announcements. *Economics Letters*, 138:5–8, 2016a.

C. Rosa. Fedspeak: Who Moves U.S. Asset Prices? *International Journal of Central Banking*, 12(4):223–261, 2016b.

P. Savor and M. Wilson. How Much Do Investors Care About Macroeconomic Risk? Evidence from Scheduled Economic Announcements. *Journal of Financial and Quantitative Analysis*, 48(2):343–375, 2013.

H. R. Stoll and R. E. Whaley. The dynamics of stock index and stock index futures returns. *Journal of Financial and Quantitative Analysis*, 25(4):441–468, 1990.

D. Vayanos. Flight to Quality, Flight to Liquidity, and the Pricing of Risk. *Working Paper*, 2004.

X. S. Zhu. Volume dynamics around FOMC announcements. *Working Paper*, 2021.

6 Tables

Table 1: Excess Liquidity Before FOMC Announcements.

		Excess Liquidity Measures (% deviation from prior-month mean)			
Statistic	Sample	Depth	Volume	Amihud	Spread
Median	Pre-FOMC	−7.0	−6.7	2.9	0.14
	Non-FOMC	1.3	0.8	3.1	0.14
	Difference	**−8.3*****	**−7.4*****	−0.2	0.00
25th percentile	Pre-FOMC	−20.5	−25.6	−12.1	−0.22
	Non-FOMC	−14.9	−18.1	−11.8	−0.14
	Difference	−5.6	−7.4	−0.3	−0.08
10th percentile	Pre-FOMC	−39.9	−47.3	−49.8	−2.57
	Non-FOMC	−32.0	−35.0	−27.4	−1.48
	Difference	−7.9	−12.4	−22.4	−1.09
High liquidity days (%)	Pre-FOMC	39.5	42.8	54.9	65.1
	Non-FOMC	52.1	50.9	55.8	69.6
	Difference	−12.6	−8.1	−1.0	−4.5

Note: This table compares excess liquidity on the trading day before an FOMC announcement with excess liquidity on non-FOMC days. Excess liquidity is the percentage deviation from the prior-month average liquidity. All measures are oriented so that positive values indicate greater liquidity. *Difference* is pre-FOMC minus non-FOMC and is reported in percentage points. The 25th and 10th percentiles describe moderately adverse and severe lower-tail liquidity conditions, respectively. High liquidity days is the percentage of days with positive excess liquidity. Statistical significance for median differences is based on two-sided permutation tests with 9,999 replications; the corresponding nonparametric bootstrap confidence intervals are used to assess whether the differences exclude zero. Bold entries denote median differences whose 95% bootstrap confidence intervals exclude zero. *** denotes significance at the 1% level. Section 2.2 defines the liquidity measures.

Table 2: Excess Liquidity and Afternoon Returns.

	Dependent variable: Afternoon return			
	Depth	Volume	Amihud	Spread
	(1)	(2)	(3)	(4)
FOMC-day intercept shift	−5.32	−5.87	−4.49	1.75
	(7.48)	(7.44)	(7.24)	(7.11)
Liquidity slope: Non-FOMC	−6.95***	−1.91*	−0.07	0.03
	(2.35)	(1.15)	(0.89)	(0.06)
Liquidity slope: FOMC	−42.97**	−23.73**	−22.95***	−2.93***
	(20.85)	(9.84)	(8.59)	(1.12)
FOMC minus non-FOMC slope	−36.02*	−21.81**	−22.88***	−2.96***
	(21.14)	(9.98)	(8.67)	(1.14)
Lagged volatility	✓	✓	✓	✓
Lagged afternoon return	✓	✓	✓	✓
Weekday fixed effects	✓	✓	✓	✓
Month fixed effects	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓
Observations	6,024	6,024	6,024	6,024
FOMC observations	215	215	215	215
Non-FOMC observations	5,809	5,809	5,809	5,809
R^2	0.032	0.031	0.031	0.033

Note: This table estimates $R_t^{PM} = \alpha + \theta FOMC_t + \beta L_{t-1} + \phi(L_{t-1} \times FOMC_t) + \delta' X_t + \varepsilon_t$ separately for each liquidity measure. Afternoon returns are expressed in basis points. The non-FOMC liquidity slope is β, the FOMC liquidity slope is $\beta + \phi$, and the reported slope difference is ϕ. Lagged excess liquidity is measured as $\ln(1 + \textit{Excess Liquidity}_{t-1})$. The sample contains designated FOMC and non-FOMC days and uses a common set of observations across columns. Controls include lagged daily volatility, the previous trading day's afternoon return, and weekday, month, and year fixed effects. Newey–West standard errors are reported in parentheses. The sample period is 1997–2024. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively, based on two-sided tests.

Table 3: Excess Liquidity and FOMC Post-announcement Returns.

	Dependent variable:			
	Post-Ann. Return			
	(1)	(2)	(3)	(4)
Intercept	-22.18	-23.34	-18.91	-9.93
	(20.80)	(21.23)	(20.58)	(19.91)
$\text{Depth}_{\text{t-1}}$	-30.59*			
	(17.36)			
$\text{DollarVolume}_{\text{t-1}}$		-16.40**		
		(8.27)		
$\text{Amihud}_{\text{t-1}}$			-18.12**	
			(7.61)	
$\text{Spread}_{\text{t-1}}$				-2.65**
				(1.04)
FFR Target Shock	-6.74***	-6.76***	-7.04***	-6.79***
	(2.25)	(2.29)	(2.28)	(2.17)
FFR Path Shock	-1.67*	-1.66*	-1.67*	-1.70*
	(0.95)	(0.96)	(0.96)	(0.94)
Prior 24-hour return	-0.18	-0.18	-0.17	-0.15
	(0.12)	(0.12)	(0.12)	(0.11)
$\text{Volatility}_{\text{t-1}}$	5.00	4.69	3.67	3.52
	(3.77)	(3.79)	(3.72)	(3.70)
MPU Index	-0.01	0.01	0.02	-0.01
	(0.11)	(0.11)	(0.11)	(0.11)
Observations	210	210	210	210
R^2	0.123	0.123	0.127	0.147
FOMC Days Only	✓	✓	✓	✓

Note: This table reports the relationship between excess liquidity and post-announcement returns on FOMC days. We estimate the following equation:

$$\begin{aligned} \textit{Post-Ann. Return}_t = \alpha + \beta \, \ln(1 + \textit{Excess Liquidity}_{t-1}) + \\ \gamma \, \textit{FFR Target Shock}_t \; + \eta \, \textit{FFR Path Shock}_t + \delta' X_t + \varepsilon_t \end{aligned}$$

where excess liquidity reflects the percentage deviation in liquidity from the average daily liquidity of the previous month. FFR path and target shocks represent shocks to the path and to the current Federal funds rate, respectively. Controls represent volatility and the 24-hour return before the announcement, along with the MPU Index level of the event. A detailed description of the variables is reported in Sections 2.2 and 2.3. Observations are related to FOMC meetings only over the period 1997–2024. Robust standard errors are reported in parentheses. Statistical significance at the 1%, 5%, and 10% levels, based on two-sided tests, is indicated by ***, **, and *, respectively.

Table 4: Announcement-Day Returns by Liquidity State.

	Dependent variable:			
	Announcement-Day Return			
	(1) Depth	(2) DollarVolume	(3) Amihud	(4) Spread
Positive Liquidity$_{t-1}$	3.86	13.34	15.65*	16.52*
	(11.35)	(10.84)	(9.33)	(9.15)
Negative Liquidity$_{t-1}$	36.63***	31.79***	34.44***	38.76***
	(9.13)	(9.41)	(9.93)	(9.96)
FFR Target Shock	-6.28***	-6.34***	-6.80***	-6.76***
	(2.29)	(2.39)	(2.38)	(2.35)
FFR Path Shock	-1.62*	-1.74*	-1.74*	-1.75*
	(0.96)	(0.98)	(0.98)	(0.98)
Pre-Ann. Return	0.81***	0.80***	0.81***	0.83***
	(0.12)	(0.13)	(0.13)	(0.13)
Volatility$_{t-1}$	6.29	5.46	4.34	3.91
	(4.16)	(4.24)	(4.08)	(4.06)
MPU Index	-0.03	-0.01	-0.01	-0.01
	(0.11)	(0.11)	(0.11)	(0.11)
Negative − Positive	32.77**	18.45	18.79	22.24
	(15.32)	(14.98)	(13.67)	(13.82)
Observations	210	210	210	210
R^2	0.383	0.373	0.374	0.375

Note: The dependent variable is the announcement-date return, from the close of FOMC$_{t-1}$ to the close of the FOMC day, in basis points, so it spans the overnight, pre-announcement, and post-announcement segments. Each column uses the excess-liquidity measure in its header to define the liquidity states. The constant is replaced by two dummies that sum to one: the positive dummy equals one when the excess-liquidity measure on FOMC$_{t-1}$ is positive, and the negative dummy equals one otherwise; the continuous excess-liquidity variable is dropped because the dummies are constructed from it. The meeting's own pre-announcement return replaces the prior 24-hour return as a control; because it is a component of the dependent variable, its coefficient is expected to be close to one. All controls are demeaned within the estimation sample, so each dummy coefficient is the fitted announcement-day return in that liquidity state at the sample mean of the controls. Standard errors are heteroskedasticity robust. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

Table 5: Post-Announcement Liquidity and Volatility Changes by Pre-Announcement Liquidity.

	Excess Depth			Excess Volume		
Statistic	Low	High	Low–High	Low	High	Low–High
Panel A: Post-Announcement Liquidity Change (%)						
Mean	247***	205***	42**	364***	297***	67**
Winsorized mean	246***	204***	41**	364***	294***	70***
Median	232***	178***	54**	340***	269***	71***
Observations	130	85	—	123	92	—
Panel B: Post-Announcement Volatility Change (%)						
Mean	176***	161***	15	174***	165***	9
Winsorized mean	176***	161***	15	174***	165***	9
Median	168***	144***	24	164***	151***	13
Observations	130	85	—	123	92	—

Note: This table reports post-announcement changes in liquidity and volatility conditional on liquidity on the trading day preceding an FOMC announcement. A post-announcement change is calculated as the percentage difference between the average value during the post-announcement interval and the average value during the final 30 minutes before the announcement. Events are classified as having low (high) pre-announcement liquidity when liquidity on $\text{FOMC}_{\text{t-1}}$ is at or below (above) its prior-month average. The Excess Depth columns use depth for this classification, whereas the Excess Volume columns use dollar volume. Low–High is the low-liquidity estimate minus the high-liquidity estimate and is therefore reported in percentage points. Winsorized means are calculated after winsorizing observations separately within each liquidity group at the 1st and 99th percentiles. Stars on the Low and High estimates indicate rejection of the null hypothesis that the corresponding statistic equals zero; stars on Low–High indicate rejection of equality between the two liquidity groups. Inference for means and winsorized means is based on two-sided Welch t-tests. Inference for individual medians is based on two-sided sign tests, and inference for differences in medians is based on two-sided bootstrap tests with 9,999 replications. All estimates are rounded to the nearest whole number. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

7 Internet Appendix

7.1 Constituent-Stock Liquidity

This section asks whether the liquidity–return relation documented in the main text reflects the liquidity of the stock market as a whole rather than that of the E-mini. Using CRSP daily data and the S&P 500 constituent list at each point in time, we construct the combined dollar volume, Amihud liquidity, and spread tightness of the index constituents, with the same excess-over-previous-month transformation and timing as the futures measures. Table A.1 shows that the constituent-based measures correlate positively with the E-mini measures, most strongly for trading activity. The return relation, however, does not carry over: the constituent-based price-impact measures do not predict post-announcement returns, and constituent dollar volume, if anything, enters with the opposite sign (Table A.2); when the E-mini measure and the corresponding aggregate enter jointly, the E-mini coefficients remain of similar magnitude and statistically significant (Table A.3). These results are consistent with the premium being tied to the cost of trading index exposure rather than single names.

Table A.1: Correlation Between E-mini and S&P 500 Constituent Excess Liquidity.

	All trading days	FOMC days
Dollar volume	0.245	0.293
Amihud liquidity	0.038	0.065
Spread tightness	0.039	0.125

Note: Pairwise correlations between each E-mini excess-liquidity measure and the same measure constructed for the S&P 500 constituents from CRSP daily data, using the S&P 500 constituent list at each point in time. All variables are excess liquidity relative to the previous month's average, measured on $\text{FOMC}_{\text{t-1}}$ and transformed as $\log(1+x)$, exactly as the futures measures. The first column uses all trading days; the second uses FOMC announcement days.

Table A.2: Post-Announcement Returns and S&P 500 Constituent Liquidity.

	Dependent variable:		
	Post-Ann. Return		
	(1) Dollar volume	(2) Amihud	(3) Spread
Intercept	-5.53	-14.52	-11.38
	(20.25)	(19.94)	(21.03)
Constituent Liquidity$_{t-1}$	114.47**	14.06	-0.15
	(53.96)	(24.00)	(0.53)
FFR Target Shock	-7.99***	-6.97***	-7.22***
	(2.42)	(2.37)	(2.43)
FFR Path Shock	-2.11**	-1.89*	-1.88**
	(0.91)	(0.96)	(0.95)
Prior 24-hour return	-0.13	-0.19	-0.18
	(0.12)	(0.12)	(0.12)
Volatility$_{t-1}$	2.66	4.69	3.78
	(3.60)	(3.94)	(3.94)
MPU Index	-0.02	-0.03	-0.02
	(0.11)	(0.11)	(0.11)
Observations	210	210	210
R^2	0.138	0.114	0.112

Note: Each column re-estimates the specification of Table 3, replacing the E-mini excess-liquidity measure with the corresponding measure constructed for the S&P 500 constituents from CRSP daily data, using the constituent list at each point in time (combined dollar volume, Amihud liquidity, and spread tightness, oriented so that higher values indicate greater liquidity; same excess-over-previous-month transformation and timing as the futures measures). The dependent variable is the post-announcement return in basis points. Standard errors are heteroskedasticity robust. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

Table A.3: Post-Announcement Returns: E-mini and Constituent Liquidity Jointly.

	Dependent variable:		
	Post-Ann. Return		
	(1) Dollar volume	(2) Amihud	(3) Spread
Intercept	-18.97	-20.43	-9.48
	(21.25)	(20.35)	(20.56)
E-mini Liquidity$_{t-1}$	-24.82***	-18.27**	-2.64**
	(8.53)	(7.50)	(1.06)
Constituent Liquidity$_{t-1}$	144.95***	14.80	-0.04
	(54.74)	(23.62)	(0.47)
FFR Target Shock	-7.92***	-7.03***	-6.85***
	(2.29)	(2.29)	(2.28)
FFR Path Shock	-1.89**	-1.72*	-1.71*
	(0.89)	(0.96)	(0.94)
Prior 24-hour return	-0.11	-0.18	-0.15
	(0.11)	(0.12)	(0.12)
Volatility$_{t-1}$	3.03	4.21	3.42
	(3.45)	(3.84)	(3.86)
MPU Index	0.03	0.02	-0.01
	(0.11)	(0.11)	(0.11)
Observations	210	210	210
R^2	0.162	0.130	0.147

Note: Each column includes both the E-mini excess-liquidity measure and the corresponding S&P 500 constituent-based measure in the specification of Table 3. Variables as defined in Table A.2. Standard errors are heteroskedasticity robust. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

7.2 Liquidity Statistics

This section reports descriptive evidence on the excess liquidity measures used in the empirical analysis. Table A.4 reports pairwise correlations among the liquidity proxies on $FOMC_{t-1}$ days and non-FOMC days. Figures A.1–A.3 summarize the behavior, classification, and dispersion of excess liquidity around FOMC announcements. These statistics show that the liquidity measures capture related, but not identical, dimensions of market liquidity.

Table A.4: Correlations Among Excess Liquidity Measures.

Panel A: $FOMC_{t-1}$ Days				
	Depth	Volume	Amihud	Spread
Depth	1.00	0.78	0.48	0.40
Volume		1.00	0.85	0.49
Amihud			1.00	0.62
Spread				1.00
Panel B: Non-FOMC Days				
	Depth	Volume	Amihud	Spread
Depth	1.00	0.65	0.31	0.35
Volume		1.00	0.86	0.33
Amihud			1.00	0.37
Spread				1.00

Note: This table reports pairwise correlations among the excess liquidity measures on $FOMC_{t-1}$ days and non-FOMC days. Non-FOMC days are trading days excluding $FOMC_{t-1}$, $FOMC_t$, and $FOMC_{t+1}$. Variables are described in Section 2.2. Each liquidity measure is expressed as percentage liquidity in excess of, or in deficiency from, its prior-month average. All measures are oriented so that higher values indicate greater liquidity.

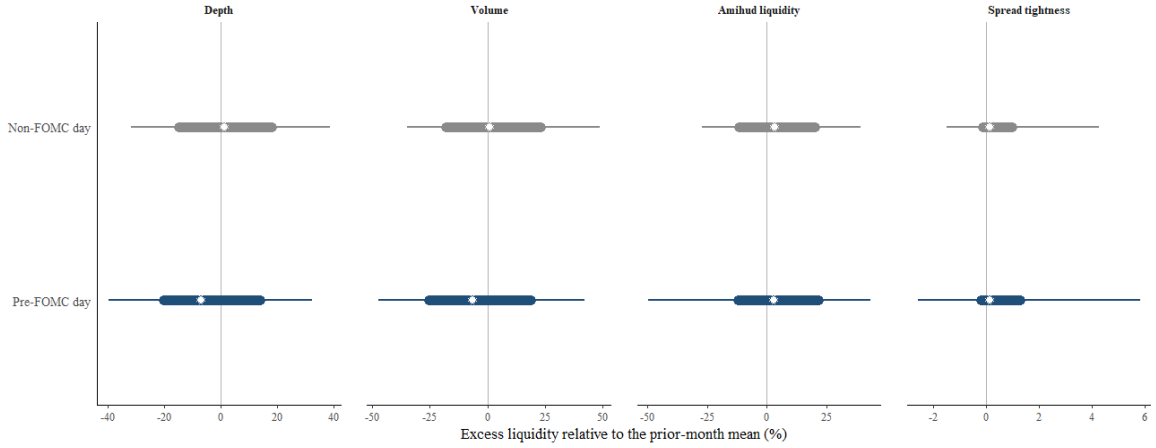


Figure A.1: Excess Liquidity Around FOMC Announcements and on non-FOMC days.

Note: This figure is the graphical counterpart to Table 1, comparing excess liquidity on $\text{FOMC}_{\text{t-1}}$ and non-FOMC days. Liquidity is scaled relative to its own average over the prior-month mean.

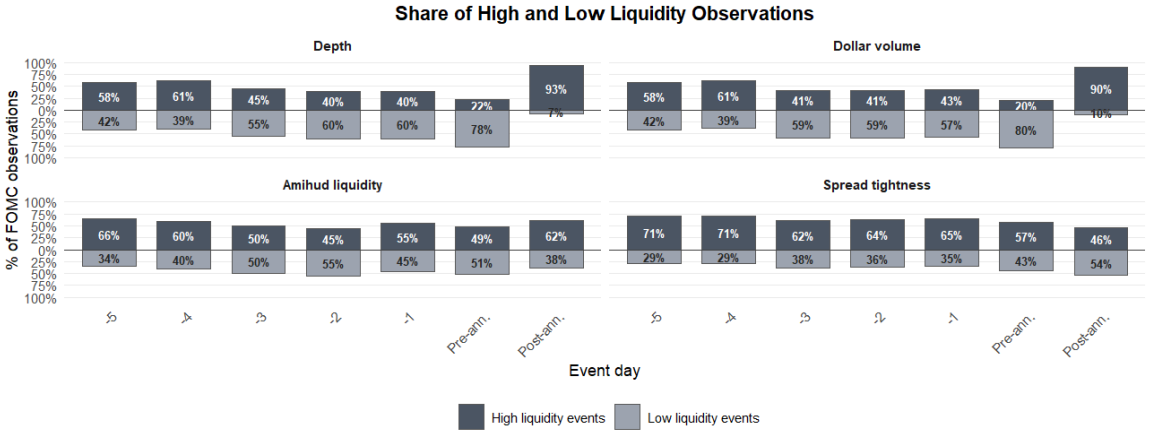


Figure A.2: Share of High- and Low-Liquidity Observations Around FOMC Announcements.

Note: This figure reports the share of observations for which each excess-liquidity measure is above or at or below its own prior month mean. Bars above the horizontal axis report high-liquidity observations, defined as excess liquidity greater than zero. Bars below the horizontal axis report low-liquidity observations, defined as excess liquidity less than or equal to zero. The daily $t-1$ shares are computed using the same $\text{FOMC}_{\text{t-1}}$ sample as Table 1. Amihud is measured using the 5-minute measure.

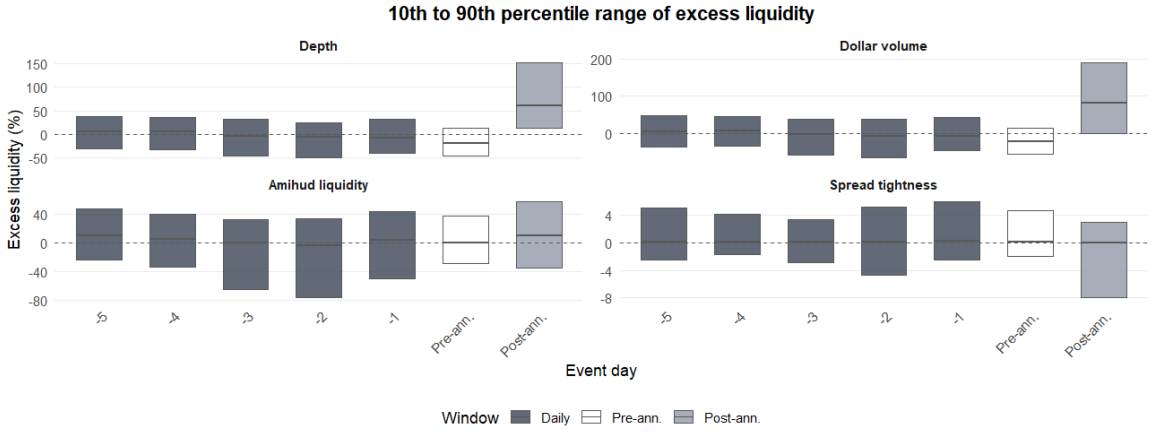


Figure A.3: Interdecile Range of Excess Liquidity Around FOMC Announcements.

Note: This figure reports the dispersion of excess liquidity around FOMC announcements. Each box spans the 10th to the 90th percentile of the excess-liquidity measure across FOMC observations, with the horizontal line inside the box marking the median. Daily liquidity is reported for event days $t-5$ through $t-1$, while pre-announcement and post-announcement liquidity are reported only for event day 0. Amihud is measured using the 5-minute measure.

7.3 Excess Liquidity and Intraday Returns

This section reports additional tests on the relation between excess liquidity and intraday returns. Table A.5 repeats the FOMC-day post-announcement return specification using the monetary policy shock measure from Jarociński and Karadi (2020). Tables A.6 and A.7 examine whether excess liquidity on the previous trading day is associated with returns during the morning window, both in the full sample of trading days and on FOMC announcement days. Tables A.8 and A.9 then test whether liquidity measured during the morning or pre-announcement window is related to returns later in the trading day.

Table A.5: Pre-FOMC Excess Liquidity and Post-announcement Returns.

	Dependent variable:			
	Post-Ann. Return			
	(1)	(2)	(3)	(4)
(Intercept)	-28.77*	-27.16*	-21.75	-16.62
	(15.95)	(15.92)	(15.40)	(15.20)
$\text{Depth}_{\text{t-1}}$	-31.68*			
	(16.72)			
$\text{Volume}_{\text{t-1}}$		-16.22**		
		(7.80)		
$\text{Amihud}_{\text{t-1}}$			-17.01**	
			(7.23)	
$\text{Spread}_{\text{t-1}}$				-2.55**
				(1.02)
JK Policy Shock	-5.56***	-5.54***	-5.68***	-5.63***
	(1.69)	(1.73)	(1.74)	(1.70)
Prior 24-hour return	-0.16	-0.16	-0.15	-0.13
	(0.11)	(0.11)	(0.11)	(0.10)
$\text{Volatility}_{\text{t-1}}$	4.93	4.64	3.67	3.51
	(3.70)	(3.71)	(3.65)	(3.64)
Observations	215	215	215	215
R^2	0.122	0.120	0.123	0.140
FOMC Days Only	✓	✓	✓	✓

Note: This table reports the relationship between excess liquidity and post-announcement returns. We estimate the following equation:

$$\textit{Post-Ann. Return}_t = \alpha + \beta \ln(1 + \textit{Excess Liquidity}_{t-1}) + \delta' X_t + \varepsilon_t$$

where excess liquidity reflects the percentage deviation in liquidity from the average daily liquidity of the previous month. Controls represent the "JK Policy Shock", the first principal component of high-frequency surprises in short-rate derivatives from Jarociński and Karadi (2020); the 24-hour return before the announcement; and volatility of $\text{FOMC}_{\text{t-1}}$. Observations are related to FOMC meetings only over the period 1997–2024. A detailed description of the variables is reported in Sections 2.2 and 2.3. Robust standard errors are reported in parentheses. Statistical significance at the 1%, 5%, and 10% levels, based on two-sided tests, is indicated by ***, **, and *, respectively.

Table A.6: Pre-FOMC Excess Liquidity and Morning Returns.

	Dependent variable: Morning return			
	Depth	Volume	Amihud	Spread
	(1)	(2)	(3)	(4)
FOMC-day intercept shift	31.51***	30.94***	28.38***	26.67***
	(5.73)	(5.75)	(5.59)	(5.74)
Liquidity slope: Non-FOMC	−5.45	−3.85**	−3.69**	−0.25**
	(4.07)	(1.85)	(1.78)	(0.11)
Liquidity slope: FOMC	39.77***	15.95**	5.85	0.63
	(11.37)	(6.51)	(5.78)	(0.61)
FOMC minus non-FOMC slope	45.21***	19.80***	9.54	0.87
	(11.74)	(6.67)	(5.97)	(0.61)
Lagged volatility	✓	✓	✓	✓
Lagged open-to-close return	✓	✓	✓	✓
Weekday fixed effects	✓	✓	✓	✓
Month fixed effects	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓
Observations	6,012	6,012	6,012	6,012
FOMC observations	215	215	215	215
Non-FOMC observations	5,797	5,797	5,797	5,797
R^2	0.014	0.014	0.014	0.014

Note: This table estimates $R_t^{AM} = \alpha + \theta FOMC_t + \beta L_{t-1} + \phi(L_{t-1} \times FOMC_t) + \delta' X_t + \varepsilon_t$ separately for each liquidity measure. Morning returns (R_t^{AM}) are expressed in basis points. On FOMC days, the morning return is measured from the previous trading day's close through the close of the minute immediately preceding the announcement. The non-FOMC liquidity slope is β, the FOMC liquidity slope is $\beta + \phi$, and the reported slope difference is ϕ. Lagged excess liquidity is measured as $\ln(1 + \textit{Excess Liquidity}_{t-1})$. The sample contains designated FOMC and non-FOMC days and uses a common set of observations across columns. Controls include lagged daily volatility, the previous trading day's open-to-close return, and weekday, month, and year fixed effects. Newey–West standard errors are reported in parentheses. The sample period is 1997–2024. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively, based on two-sided tests.

Table A.7: Pre-FOMC Excess Liquidity and Pre-announcement Returns.

	Dependent variable:			
	Pre-Ann. Return			
	(1)	(2)	(3)	(4)
Intercept	-11.30	-10.74	-13.24	-17.69*
	(11.20)	(11.09)	(10.60)	(10.24)
$\text{Depth}_{\text{t-1}}$	19.58*			
	(11.25)			
$\text{Volume}_{\text{t-1}}$		9.99		
		(6.35)		
$\text{Amihud}_{\text{t-1}}$			10.95*	
			(6.61)	
$\text{Spread}_{\text{t-1}}$				1.01
				(0.68)
Daily $\text{Volatility}_{\text{t-1}}$	11.08***	11.30***	11.82***	11.67***
	(2.38)	(2.33)	(2.29)	(2.33)
Previous-day Return	-0.02	-0.02	-0.03	-0.04
	(0.07)	(0.07)	(0.08)	(0.08)
MPU Index	-0.11*	-0.12**	-0.13**	-0.11*
	(0.06)	(0.06)	(0.06)	(0.06)
Observations	210	210	210	210
R^2	0.249	0.248	0.251	0.249
FOMC Days Only	✓	✓	✓	✓

Note: This table reports the relationship between excess liquidity on the trading day preceding an FOMC announcement and the pre-announcement return. For each liquidity measure, we estimate

$$\textit{Pre-Ann. Return}_t = \alpha + \beta \ln(1 + \textit{Excess Liquidity}_{t-1}) + \gamma \textit{Volatility}_{t-1} + \rho \textit{Return}_{t-1} + \eta MPU_t + \varepsilon_t.$$

Excess liquidity is the percentage deviation of liquidity from its average daily level over the preceding month. The previous-day return is the open-to-close return on the trading day preceding the FOMC announcement. The MPU Index measures monetary-policy uncertainty on the announcement day. Pre-announcement and previous-day returns are expressed in basis points. The sample contains scheduled FOMC announcement days from 1997 through 2024. A detailed description of the variables is provided in Sections 2.2 and 2.3. Heteroskedasticity-robust standard errors are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively, based on two-sided tests.

Table A.8: Morning Excess Liquidity and Afternoon Returns.

	Dependent variable: Afternoon return			
	Depth	Volume	Amihud	Spread
	(1)	(2)	(3)	(4)
FOMC-day intercept shift	−4.50	−4.47	−3.22	1.47
	(8.28)	(8.00)	(7.39)	(7.08)
Liquidity slope: Non-FOMC	−6.71***	−1.75	0.01	0.04
	(2.16)	(1.08)	(0.84)	(0.07)
Liquidity slope: FOMC	−14.42	−8.63	−13.83**	−2.41**
	(19.42)	(7.93)	(7.00)	(1.10)
FOMC minus non-FOMC slope	−7.71	−6.88	−13.84**	−2.46**
	(19.74)	(8.12)	(7.02)	(1.12)
Lagged volatility	✓	✓	✓	✓
Previous day open-to-close return	✓	✓	✓	✓
Weekday fixed effects	✓	✓	✓	✓
Month fixed effects	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓
Observations	6,024	6,024	6,024	6,024
FOMC observations	215	215	215	215
Non-FOMC observations	5,809	5,809	5,809	5,809
R^2	0.032	0.031	0.031	0.033

Note:

Note: This table estimates $R_t^{PM} = \alpha + \theta FOMC_t + \beta L_t^{pre} + \phi(L_t^{pre} \times FOMC_t) + \delta' X_t + \varepsilon_t$ separately for each liquidity measure. Afternoon returns (R_t^{PM}) are expressed in basis points. On FOMC days, the afternoon return is the post-announcement return. The non-FOMC liquidity slope is β, the FOMC liquidity slope is $\beta+\phi$, and the reported slope difference is ϕ. Pre-announcement excess liquidity, L_t^{pre}, is measured during the morning window and transformed to reduce the influence of extreme observations. The sample contains designated FOMC and non-FOMC days and uses a common set of observations across columns. Controls include lagged daily volatility, the previous trading day's open-to-close return, and weekday, month, and year fixed effects. Newey–West standard errors are reported in parentheses. The sample period is 1997–2024. Statistical significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *, respectively.

Table A.9: Morning Excess Liquidity and Post-announcement Returns.

	Dependent variable:			
	Post-Ann. Return			
	(1)	(2)	(3)	(4)
Constant	-13.59	-15.26	-17.09	-9.23
	(21.83)	(22.21)	(20.92)	(19.97)
$\text{Depth}_{\text{pre-ann.}}$	-9.51			
	(20.37)			
$\text{Volume}_{\text{pre-ann.}}$		-6.42		
		(8.82)		
$\text{Amihud}_{\text{pre-ann.}}$			-14.77*	
			(8.59)	
$\text{Spread}_{\text{pre-ann.}}$				-2.87**
				(1.28)
FFR target shock	-6.78***	-6.80***	-6.98***	-7.03***
	(2.37)	(2.37)	(2.33)	(2.25)
FFR path shock	-1.92**	-1.90*	-1.89*	-1.94**
	(0.96)	(0.97)	(0.97)	(0.94)
$\text{Volatility}_{\text{t-1}}$	2.35	2.38	2.19	2.27
	(3.50)	(3.48)	(3.40)	(3.42)
Previous-day Return	-0.05	-0.05	-0.03	-0.01
	(0.10)	(0.10)	(0.10)	(0.10)
MPU Index	-0.01	0.00	0.03	0.01
	(0.11)	(0.11)	(0.11)	(0.11)
Observations	210	210	210	210
R^2	0.094	0.095	0.102	0.126
FOMC Days Only	✓	✓	✓	✓

Note: This table reports the relationship between pre-announcement (morning) excess liquidity and post-announcement returns on FOMC days. We estimate the following equation:

$$\textit{Post-Ann. Return}_t = \alpha + \beta \ln(1 + \textit{Excess Liquidity}_{Pre-ann.}) + \gamma\, \textit{FFR Target Shock}_t + \eta\, \textit{FFR Path Shock}_t + \delta' X_t + \varepsilon_t$$

where excess liquidity reflects the percentage deviation in liquidity from the average morning liquidity of the previous month. FFR path and target shocks represent shocks to the path and to the current Federal funds rate, respectively. X_t represent volatility and open to close return of $\text{FOMC}_{\text{t-1}}$, along with the MPU Index level of the event. Observations are related to FOMC meetings only over the period 1997–2024. A detailed description of the variables is reported in Sections 2.2 and 2.3. Robust standard errors are reported in parentheses. Statistical significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *, respectively.

7.4 Alternative Definitions of the Announcement-Day Return

Tables A.10 to A.13 repeat the specification of Table 4 under alternative definitions of the dependent variable and of the liquidity states. Table A.10 uses the post-announcement return, for which the negative average is concentrated after high-liquidity days; Table A.11 uses the adjusted post-announcement return, which adds back the average pre-announcement premium of 27.75 basis points; and Tables A.12 and A.13 replace the binary split with liquidity terciles, for the adjusted and the announcement-day returns, respectively. The negative-minus-positive differential for depth is 33 to 34 basis points across versions, and the tercile results show that the pattern is concentrated in the top liquidity tercile rather than an artifact of the binary split.

Table A.10: Post-Announcement Returns by Liquidity State.

	Dependent variable:			
	Post-Ann. Return			
	(1) Depth	(2) DollarVolume	(3) Amihud	(4) Spread
Positive Liquidity$_{t-1}$	-22.88**	-13.50	-10.01	-10.02
	(11.26)	(10.62)	(9.21)	(8.95)
Negative Liquidity$_{t-1}$	10.72	6.00	7.51	13.14
	(8.96)	(9.30)	(9.80)	(9.99)
FFR Target Shock	-6.53***	-6.59***	-7.06***	-6.98***
	(2.30)	(2.39)	(2.40)	(2.36)
FFR Path Shock	-1.62*	-1.75*	-1.76*	-1.75*
	(0.94)	(0.96)	(0.96)	(0.96)
Prior 24-hour return	-0.19	-0.19	-0.18	-0.18
	(0.12)	(0.12)	(0.12)	(0.12)
Volatility$_{t-1}$	5.95	5.08	3.89	3.62
	(3.81)	(3.85)	(3.75)	(3.71)
MPU Index	-0.03	0.00	-0.00	-0.00
	(0.11)	(0.11)	(0.11)	(0.11)
Negative − Positive	33.60**	19.50	17.52	23.16*
	(15.07)	(14.68)	(13.43)	(13.64)
Observations	210	210	210	210
R^2	0.134	0.120	0.119	0.123

Note: As in Table 4, but the dependent variable is the post-announcement return and the prior 24-hour return is the return control, as in Table 3. Each dummy coefficient is the fitted post-announcement return, in basis points, in that liquidity state at the sample mean of the controls. Standard errors are heteroskedasticity robust. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

Table A.11: Adjusted Post-Announcement Returns by Liquidity State.

	Dependent variable:			
	Adjusted Post-Ann. Return			
	(1) Depth	(2) DollarVolume	(3) Amihud	(4) Spread
Positive Liquidity$_{t-1}$	4.86	14.24	17.74*	17.72**
	(11.26)	(10.62)	(9.21)	(8.95)
Negative Liquidity$_{t-1}$	38.47***	33.74***	35.26***	40.88***
	(8.96)	(9.30)	(9.80)	(9.99)
FFR Target Shock	-6.53***	-6.59***	-7.06***	-6.98***
	(2.30)	(2.39)	(2.40)	(2.36)
FFR Path Shock	-1.62*	-1.75*	-1.76*	-1.75*
	(0.94)	(0.96)	(0.96)	(0.96)
Prior 24-hour return	-0.19	-0.19	-0.18	-0.18
	(0.12)	(0.12)	(0.12)	(0.12)
Volatility$_{t-1}$	5.95	5.08	3.89	3.62
	(3.81)	(3.85)	(3.75)	(3.71)
MPU Index	-0.03	0.00	-0.00	-0.00
	(0.11)	(0.11)	(0.11)	(0.11)
Negative − Positive	33.60**	19.50	17.52	23.16*
	(15.07)	(14.68)	(13.43)	(13.64)
Observations	210	210	210	210
R^2	0.186	0.172	0.171	0.175

Note: The dependent variable is the adjusted post-announcement return: the post-announcement return plus the average pre-announcement return across all 215 meetings (+27.75 bp, $t = 5.72$). Because the adjustment is a constant, the differences between states and all control coefficients are identical to Table A.10; the levels can be read against the full average announcement premium. The adjustment treats the average pre-announcement return as a known constant; the specification in Table 4 avoids this assumption and reaches the same conclusion. Standard errors are heteroskedasticity robust. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

Table A.12: Adjusted Post-Announcement Returns by Liquidity Tercile.

	Dependent variable:			
	Adjusted Post-Ann. Return			
	(1) Depth	(2) DollarVolume	(3) Amihud	(4) Spread
Low Liquidity$_{t-1}$ (bottom tercile)	37.29***	37.87***	33.15***	38.59***
	(12.49)	(12.83)	(11.70)	(9.81)
Middle Liquidity$_{t-1}$ (middle tercile)	35.84***	34.50***	12.55	43.67***
	(11.95)	(10.70)	(12.15)	(9.82)
High Liquidity$_{t-1}$ (top tercile)	3.86	4.62	31.29***	-5.27
	(11.33)	(12.30)	(10.88)	(14.29)
FFR Target Shock	-6.44***	-6.43***	-7.13***	-6.45***
	(2.31)	(2.36)	(2.42)	(2.24)
FFR Path Shock	-1.58*	-1.62*	-1.76*	-1.72*
	(0.93)	(0.96)	(0.97)	(0.94)
Prior 24-hour return	-0.19	-0.20	-0.19	-0.17
	(0.12)	(0.12)	(0.12)	(0.11)
Volatility$_{t-1}$	6.15	5.87	4.16	4.86
	(3.86)	(3.84)	(3.75)	(3.58)
MPU Index	-0.02	-0.01	-0.02	-0.04
	(0.11)	(0.11)	(0.11)	(0.11)
Low − High	33.43*	33.25*	1.86	43.86**
	(18.03)	(18.53)	(15.83)	(17.85)
Observations	210	210	210	210
R^2	0.182	0.182	0.172	0.207

Note: The dependent variable is the adjusted post-announcement return, as in Table A.11: the post-announcement return plus the average pre-announcement return across all 215 meetings (+27.75 bp, $t = 5.72$). The binary liquidity states are replaced by three dummies that sum to one, indicating the bottom, middle, and top tercile of the excess-liquidity measure on FOMC$_{t-1}$, computed within the estimation sample ($\approx$ 70 meetings per tercile). All controls are demeaned within the estimation sample. Standard errors are heteroskedasticity robust. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

Table A.13: Announcement-Day Returns by Liquidity Tercile.

	Dependent variable:			
	Announcement-Day Return			
	(1) Depth	(2) DollarVolume	(3) Amihud	(4) Spread
Low Liquidity$_{t-1}$ (bottom tercile)	36.03***	34.97***	31.75***	35.88***
	(12.62)	(12.98)	(11.72)	(9.77)
Middle Liquidity$_{t-1}$ (middle tercile)	32.85***	33.81***	11.23	45.40***
	(12.35)	(10.78)	(12.52)	(10.20)
High Liquidity$_{t-1}$ (top tercile)	3.56	3.66	29.46***	-8.84
	(11.22)	(12.60)	(10.77)	(14.46)
FFR Target Shock	-6.22***	-6.17***	-6.88***	-6.13***
	(2.29)	(2.36)	(2.40)	(2.21)
FFR Path Shock	-1.59*	-1.63*	-1.76*	-1.70*
	(0.95)	(0.98)	(0.99)	(0.97)
Pre-Ann. Return	0.82***	0.80***	0.82***	0.80***
	(0.12)	(0.13)	(0.13)	(0.12)
Volatility$_{t-1}$	6.36	6.29	4.45	5.78
	(4.23)	(4.23)	(4.07)	(3.94)
MPU Index	-0.03	-0.02	-0.03	-0.05
	(0.11)	(0.11)	(0.11)	(0.11)
Low − High	32.47*	31.31*	2.30	44.72**
	(18.11)	(18.88)	(15.73)	(17.99)
Observations	210	210	210	210
R^2	0.380	0.380	0.374	0.405

Note: As in Table 4, with the binary split replaced by three dummies that sum to one, indicating the bottom, middle, and top tercile of the excess-liquidity measure on FOMC$_{t-1}$, computed within the estimation sample ($\approx$ 70 meetings per tercile). Standard errors are heteroskedasticity robust. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

7.5 Post-Announcement Liquidity Changes and the Direction of the News

Table A.14 complements Table 5 by reporting the post-announcement changes in depth and dollar volume separately by the sign of the 24-hour pre-announcement return and by the sign of the JK policy shock.

Table A.14: Post-Announcement Depth and Volume Changes by the Signs of the Pre-Announcement Return and JK Shock.

	Depth Change			Dollar-Volume Change		
Statistic	Nonpositive	Positive	Diff.	Nonpositive	Positive	Diff.
Panel A: Conditioning Variable: Pre-Announcement Return						
Mean	231***	230***	1	299***	355***	-56**
Winsorized mean	231***	229***	2	299***	353***	-54**
Median	201***	213***	-11	283***	324***	-41
Observations	74	141	—	74	141	—
Panel B: Conditioning Variable: JK Shock						
Mean	231***	230***	1	345***	325***	20
Winsorized mean	229***	228***	0	345***	323***	22
Median	209***	206***	2	335***	281***	54*
Observations	111	104	—	111	104	—

Note: This table reports post-announcement changes in depth and dollar volume conditional on the signs of the pre-announcement return and the JK shock. A post-announcement change is calculated as the percentage difference between the average value during the post-announcement interval and the average value during the final 30 minutes before the announcement. Panel A classifies FOMC announcements according to the sign of the pre-announcement return (Drift), while Panel B classifies announcements according to the sign of the JK shock. Nonpositive includes observations for which the corresponding conditioning variable is less than or equal to zero; Positive includes observations for which it is greater than zero. Nonpos.–Pos. reports the nonpositive-group estimate minus the positive-group estimate and is expressed in percentage points.
Winsorized means are calculated after winsorizing observations separately within each group at the 1st and 99th percentiles. Stars on the group estimates test whether the corresponding statistic equals zero; stars on the differences test equality between the two groups. Inference for means and winsorized means is based on two-sided Welch t-tests. Inference for individual medians is based on two-sided sign tests, and inference for differences in medians is based on two-sided bootstrap tests with 9,999 replications. All estimates are reported in percent and rounded to the nearest whole number. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

7.6 Residual Liquidity

This section presents the relation between excess liquidity and post-announcement returns in a two-step form. Table A.15 first estimates the component of pre-FOMC excess liquidity that is associated with predetermined market variables, including the VIX observed on the previous Friday, lagged volatility, lagged returns, and the MPU Index. Table A.16 then re-estimates the return regression using the residuals from these first-stage regressions. Because the second stage includes the same predetermined variables as controls, the residual specification is numerically equivalent to including excess liquidity directly alongside those controls. We report it as a transparent way of showing how much of the variation in pre-FOMC liquidity is explained by prevailing market conditions, and how the return relation reads once that component is stripped out.

Table A.15: Predictable Components of Pre-FOMC Excess Liquidity.

	Dependent variable:			
	Depth	Volume	Amihud	Spread
	(1)	(2)	(3)	(4)
Constant	-0.06	-0.34*	-0.43**	-3.27
	(0.10)	(0.19)	(0.20)	(2.21)
$\text{VIX}_{\text{previous Friday}}$	-0.02***	-0.03**	0.01	0.37**
	(0.01)	(0.01)	(0.01)	(0.16)
$\text{Volatility}_{\text{t-1}}$	0.07***	0.09***	-0.04	-0.84**
	(0.02)	(0.03)	(0.03)	(0.36)
Previous-day Return	0.00	0.00	0.00	0.01
	(0.00)	(0.00)	(0.00)	(0.01)
MPU Index	0.00	0.00***	0.00***	0.01
	(0.00)	(0.00)	(0.00)	(0.01)
Observations	210	210	210	210
R^2	0.120	0.087	0.057	0.082
Pre-FOMC Days Only	✓	✓	✓	✓

Note: This table reports the relationship between excess liquidity of the day before the announcement and market variables. We estimate the following equation:

$$Excess\ Liquidity_{FOMCt-1} = \alpha + \beta\, VIX_{Previous\ Friday} + \gamma\, Volatility_{FOMCt-2} + \eta\, Return_{FOMCt-2} + \zeta\, MPU + \varepsilon_t$$

where excess liquidity reflects the percentage deviation in liquidity from the average daily liquidity of the previous month. $\text{VIX}_{\text{Previous Friday}}$ represents the level of VIX of the last trading day of the week prior to FOMC announcements. $\text{Return}_{\text{t-1}}$ represents the open to close return of $\text{FOMC}_{\text{t-1}}$. Observations are related to FOMC meetings only over the period 1997–2024. A detailed description of the variables is reported in Sections 2.2 and 2.3. Robust standard errors are reported in parentheses. Statistical significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *, respectively.

Table A.16: Liquidity Residuals and Post-announcement Returns.

	Dependent variable:			
	Post-Ann. Return			
	(1)	(2)	(3)	(4)
Constant	29.76	29.46	29.77	28.15
	(24.91)	(25.06)	(25.02)	(24.98)
Residual $\text{Depth}_{\text{t-1}}$	-46.34***			
	(17.59)			
Residual $\text{Volume}_{\text{t-1}}$		-21.07**		
		(8.54)		
Residual $\text{Amihud}_{\text{t-1}}$			-18.36**	
			(8.04)	
Residual $\text{Spread}_{\text{t-1}}$				-2.48**
				(1.19)
FFR target shock	-8.31***	-8.20***	-8.29***	-7.72***
	(2.32)	(2.38)	(2.40)	(2.36)
FFR path shock	-1.93**	-1.93**	-1.95**	-1.93**
	(0.91)	(0.93)	(0.93)	(0.93)
$\text{VIX}_{\text{Previous Friday}}$	-3.80**	-3.77*	-3.80*	-3.65*
	(1.92)	(1.95)	(1.97)	(2.01)
$\text{Volatility}_{\text{t-1}}$	10.06*	10.00*	10.06*	9.74
	(5.67)	(5.79)	(5.88)	(6.08)
Previous-day Return	0.01	0.01	0.01	0.00
	(0.09)	(0.10)	(0.10)	(0.10)
MPU Index	-0.04	-0.04	-0.04	-0.03
	(0.11)	(0.11)	(0.11)	(0.11)
Observations	210	210	210	210
R^2	0.143	0.136	0.133	0.145
FOMC Days Only	✓	✓	✓	✓

Note: This table reports the relationship between excess liquidity residuals, estimated from Table A.15, and post-announcement returns. We estimate the following equation:

$$\begin{aligned} \textit{Post-Ann. Return}_t = \alpha + \beta\, \textit{Excess Liquidity Residuals}_{t-1} + \\ \gamma\, \textit{FFR Target Shock}_t + \eta\, \textit{FFR Path Shock}_t + \delta' X_t + \varepsilon_t \end{aligned}$$

where excess liquidity reflects the percentage deviation in liquidity from the average daily liquidity of the previous month. FFR path and target shocks represent shocks to the path and to the current Federal funds rate, respectively. Controls represent the level of VIX of the last trading day of the week prior to the announcement, volatility and open to close return of $\text{FOMC}_{\text{t-1}}$, and the MPU Index level of the event. Observations are related to FOMC meetings only over the period 1997–2024. A detailed description of the variables is reported in Sections 2.2 and 2.3. Robust standard errors are reported in parentheses. Statistical significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *, respectively.

7.7 Subsample Analysis

This section examines whether the main liquidity-return relation is also present in the more recent part of the sample. Table A.17 repeats the baseline specifications over the 2015–2024 period, excluding observations from 2020 and 2021 to reduce the influence of COVID-related market outliers.

Table A.17: Subsample Analysis: Excess Liquidity and Afternoon Returns.

	Dependent variable: Afternoon return			
	Depth	Volume	Amihud	Spread
	(1)	(2)	(3)	(4)
FOMC-day intercept shift	−21.51	−17.65	−15.99	−15.17
	(13.48)	(12.89)	(12.94)	(13.06)
Liquidity slope: Non-FOMC	−1.97	−0.58	0.76	0.35*
	(2.80)	(1.32)	(0.99)	(0.18)
Liquidity slope: FOMC	−77.36***	−53.15***	−31.10*	−1.96
	(29.71)	(16.69)	(17.26)	(1.59)
FOMC minus non-FOMC slope	−75.39**	−52.58***	−31.86*	−2.32
	(30.35)	(16.56)	(17.19)	(1.61)
Lagged volatility	✓	✓	✓	✓
Lagged afternoon return	✓	✓	✓	✓
Weekday fixed effects	✓	✓	✓	✓
Month fixed effects	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓
Observations	1,806	1,806	1,806	1,806
FOMC observations	65	65	65	65
Non-FOMC observations	1,741	1,741	1,741	1,741
R^2	0.030	0.031	0.024	0.022

Note: This table estimates $R_t^{PM} = \alpha + \theta FOMC_t + \beta L_{t-1} + \phi(L_{t-1} \times FOMC_t) + \delta' X_t + \varepsilon_t$ separately for each liquidity measure. Afternoon returns (R_t^{PM}) are expressed in basis points. On FOMC days, the afternoon return is the post-announcement return. The non-FOMC liquidity slope is β, the FOMC liquidity slope is $\beta+\phi$, and the reported slope difference is ϕ. Lagged excess liquidity, L_{t-1}, is measured on the previous trading day and transformed to reduce the influence of extreme observations. The sample contains designated FOMC and non-FOMC days and uses a common set of observations across columns. Controls include lagged daily volatility, the previous trading day's open-to-close return, and weekday, month, and year fixed effects. Newey–West standard errors with 22 lags are reported in parentheses. The sample starts in 2015 and excludes February 2020 through December 2021 to reduce the influence of COVID-related market outliers. Statistical significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *, respectively.

7.8 FOMC Liquidity Premium Conditional on Recent Macroeconomic News

To examine how the liquidity premium relates to recent macroeconomic news, we consider seven indicators: consumer confidence, CPI, initial jobless claims, ISM manufacturing, new home sales, personal spending, and retail sales. For each FOMC announcement and each indicator, we use the most recent release available before the announcement and classify its surprise as positive, negative, or zero. A surprise is positive (negative) when the actual value exceeds (falls below) the survey forecast. When the actual value equals the survey forecast, we classify the release as having no surprise.[17] Because only CPI and personal spending have a sufficient number of zero-surprise observations, we report the no-surprise category only for these two indicators. In our sample, macroeconomic releases occurring on the same day as an FOMC announcement are always released before the FOMC announcement.

Table A.18 reports the FOMC announcement liquidity premium (post-announcement return given low liquidity minus post-announcement return given high liquidity) conditional on the surprise in the most recent release of each indicator. The average liquidity premium is positive in every reported subsample and statistically significant in eleven of the sixteen subsamples. Thus, the FOMC-day liquidity premium is present across different recent macroeconomic news states rather than being concentrated exclusively in meetings preceded by either positive or negative surprises. The premium is particularly large following a positive surprise in initial jobless claims, a positive CPI surprise, and a negative surprise in ISM manufacturing.

We obtain similar results when conditioning the liquidity premium on positive and negative surprises in durable goods orders, housing starts, industrial production, ISM services, consumer sentiment, and the unemployment rate.

[17] A positive (negative) surprise does not necessarily represent positive (negative) news for the economy. For example, a positive CPI surprise means that inflation was higher than expected and may therefore constitute negative economic news. By contrast, a positive consumer confidence surprise indicates that consumers' assessment of economic conditions was more favorable than expected.

Table A.18: Liquidity Premium Conditioned on Recent Macroeconomic News

Macroeconomic News	Surprise	Events	Liquidity Premium
Consumer Confidence	Positive	98	10.77*
Consumer Confidence	Negative	76	18.64**
CPI	Positive	46	19.73*
CPI	Negative	59	7.62
CPI	No Surprise	72	18.18**
Initial Jobless Claims	Positive	81	26.07***
Initial Jobless Claims	Negative	93	5.08
ISM Manufacturing	Positive	85	8.79
ISM Manufacturing	Negative	87	18.72**
New Home Sales	Positive	78	10.90
New Home Sales	Negative	96	19.07**
Personal Spending	Positive	62	2.95
Personal Spending	Negative	56	18.60*
Personal Spending	No Surprise	59	24.44**
Retail Sales	Positive	75	22.34**
Retail Sales	Negative	81	17.66**

Note: This table reports the realized liquidity premium conditional on the most recent surprise in each listed macroeconomic release before an FOMC meeting. Meetings are classified according to whether the release produced a positive, negative, or zero surprise relative to market expectations. "Events" is the number of FOMC meetings in the corresponding subsample. "Liquidity Premium" is the average post-announcement return, in basis points, conditioned on the previous ex-ante level of liquidity (low minus high liquidity). Statistical significance at the 1%, 5%, and 10% levels is denoted by ***, **, and *, respectively.

7.9 Alternative Sample Definitions

This section re-estimates the specification of Table 3 under two alternative sample definitions. Table A.19 drops the eight meetings whose policy statement was released at 12:30 p.m., for which the post-announcement window is roughly three and a half hours instead of two. Table A.20 includes the October 2008 meeting, which the baseline sample excludes because its 24-hour pre-announcement return exceeds 10%.

Table A.19: Excess Liquidity and Post-Announcement Returns: Excluding 12:30 p.m. Releases.

	Dependent variable: Post-Ann. Return			
	(1)	(2)	(3)	(4)
Intercept	-29.51	-30.46	-25.89	-15.56
	(21.36)	(21.97)	(21.41)	(20.62)
$\text{Depth}_{\text{t-1}}$	-33.92*			
	(17.22)			
$\text{DollarVolume}_{\text{t-1}}$		-17.47**		
		(8.27)		
$\text{Amihud}_{\text{t-1}}$			-19.06**	
			(7.70)	
$\text{Spread}_{\text{t-1}}$				-2.62**
				(1.04)
FFR Target Shock	-6.80***	-6.83***	-7.13***	-6.87***
	(2.23)	(2.27)	(2.27)	(2.16)
FFR Path Shock	-1.65*	-1.65*	-1.67*	-1.70*
	(0.95)	(0.96)	(0.96)	(0.95)
Prior 24-hour return	-0.17	-0.17	-0.16	-0.14
	(0.12)	(0.12)	(0.12)	(0.11)
$\text{Volatility}_{\text{t-1}}$	5.45	5.10	4.03	3.88
	(3.88)	(3.91)	(3.85)	(3.83)
MPU Index	0.01	0.03	0.04	0.01
	(0.11)	(0.11)	(0.11)	(0.11)
Observations	202	202	202	202
R^2	0.130	0.129	0.133	0.150

Note: This table re-estimates the specification of Table 3 excluding the eight FOMC meetings whose policy statement was released at 12:30 p.m. (April, June, and November 2011; January, April, June, September, and December 2012). All variables are as in Table 3. Robust standard errors are reported in parentheses. Statistical significance at the 1%, 5%, and 10% levels, based on two-sided tests, is indicated by ***, **, and *, respectively.

Table A.20: Excess Liquidity and Post-Announcement Returns: Including the October 2008 Meeting.

	Dependent variable: Post-Ann. Return			
	(1)	(2)	(3)	(4)
Intercept	-14.37	-16.23	-12.55	-3.43
	(20.37)	(20.76)	(20.08)	(19.47)
$\text{Depth}_{\text{t-1}}$	-26.66			
	(17.94)			
$\text{DollarVolume}_{\text{t-1}}$		-15.30*		
		(8.55)		
$\text{Amihud}_{\text{t-1}}$			-17.79**	
			(7.69)	
$\text{Spread}_{\text{t-1}}$				-2.55**
				(1.02)
FFR Target Shock	-6.59***	-6.60***	-6.87***	-6.62***
	(2.39)	(2.41)	(2.41)	(2.30)
FFR Path Shock	-1.62*	-1.60*	-1.61*	-1.64*
	(0.93)	(0.94)	(0.94)	(0.93)
Prior 24-hour return	-0.27**	-0.27**	-0.25**	-0.24**
	(0.11)	(0.11)	(0.11)	(0.11)
$\text{Volatility}_{\text{t-1}}$	3.87	3.66	2.72	2.52
	(3.82)	(3.82)	(3.75)	(3.72)
MPU Index	-0.01	0.01	0.02	-0.01
	(0.11)	(0.11)	(0.11)	(0.11)
Observations	211	211	211	211
R^2	0.136	0.137	0.142	0.159

Note: This table re-estimates the specification of Table 3 including the October 29, 2008 FOMC meeting, which the baseline sample excludes because its 24-hour pre-announcement return exceeds 10%. All variables are as in Table 3. Robust standard errors are reported in parentheses. Statistical significance at the 1%, 5%, and 10% levels, based on two-sided tests, is indicated by ***, **, and *, respectively.